



## Review

## Advances in photodetectors via photogating effect

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## ABSTRACT

Photodetectors, essential for converting optical signals into electrical signals, are widely used in communication, imaging, and sensing. The photogating effect, where trapped photogenerated carriers influence carrier transport, has gained attention for enhancing detector performance. By modulating carrier transport, it significantly improves sensitivity and dynamic range. The photogating effect plays a key role in determining parameters such as gain, responsivity, and specific detectivity by adjusting the Fermi level and conductivity. Here, we detail the application of the photogating effect in photodetectors and their impact on performance. The paper discusses the principles behind the photogating effect and its influence on the performance of different material systems, such as low-dimensional materials and perovskite materials-based photodetector. In particular, low-dimensional materials and perovskite materials, with their unique optoelectronic properties, can effectively induce the photogating effect through the construction of heterojunctions or the introduction of trap states, thereby achieving high-performance photodetectors.

## 1. Introduction

The development of photodetectors can be traced back to the 1940s, with the primary forms being photodiodes and photomultiplier tubes [1]. At that time, photodetectors relied on semiconductor materials (such as Si and Ge) and vacuum technologies, and were widely applied in early detection and imaging [2,3]. With the continuous advancements in semiconductor processing and integrated circuit technologies, photodetectors began to see broad commercial adoption in the 1960s [4]. For instance, Si photodiodes rapidly developed for use in communication and sensor applications [5]. Furthermore, introducing photoconductive sensors and optical fiber technologies further promoted the diversification of photodetector applications [6]. In recent years, photodetector technology has entered a new stage of development, especially those based on organic materials, quantum dots, low-dimensional materials, and perovskite materials, which have gradually become research hotspots [7–9]. These novel materials not only exhibit excellent optoelectronic properties but also feature tunable band gaps, allowing them to operate across a broader wavelength range. This has expanded the application scope of photodetectors, covering fields ranging from traditional optical communication and imaging to

emerging areas such as wearable devices and smart monitoring systems [10–14].

Photodetectors are commonly classified based on the spectral range they detect: (I) Ultraviolet photodetectors operate in the 10–400 nm range and are typically used in applications such as flame detection, environmental monitoring, and space exploration [15]; (II) Visible photodetectors detect light in the 400–760 nm range and find use in imaging, displays, and visible light communication (VLC) [1]; (III) Infrared photodetectors respond to wavelengths beyond 760 nm, covering near-infrared (NIR), mid-infrared (MIR), and even far-infrared (FIR) ranges. These are crucial for applications in optical fiber communication, biomedical imaging, night vision, and thermal sensing [7].

Regardless of the wavelength range of light signals detected by photodetectors, the mechanism by which they convert light signals into electrical signals is fundamentally the same [16–18]. This mechanism is typically the result of the synergistic effect of several phenomena: (a) Photovoltaic effect: Under illumination, photons are absorbed by the material, generating photoexcited electron-hole pairs. These pairs are separated by the internal electric field, resulting in the generation of photocurrent [19]; (b) Photoconductive effect: Photoconductivity refers

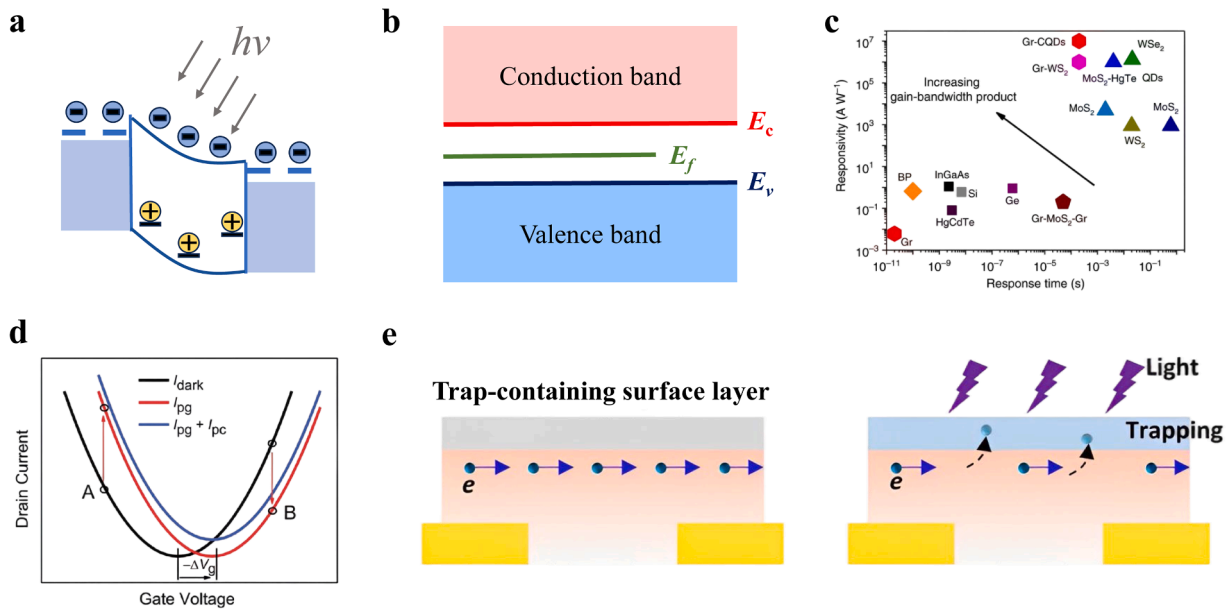
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**Fig. 1.** (a) Schematic diagram of the photogating effect, showing the modulation of electron transport by hole trap states; (b) The relative positions of  $E_c$ ,  $E_f$ , and  $E_v$ ; (c) Relationship between the responsivity and response time of common photodetectors [44]; (d) Transfer characteristics curve: the black curve represents dark current, the red curve shows the shift due to the photogating effect under illumination, and the blue curve illustrates the actual shift with both horizontal and vertical displacements [23]; (e) Schematic of the NPC phenomenon caused by photoexcited carriers being trapped by trap states [45].

to the change in the conductivity of optoelectronic devices caused by the generation of non-equilibrium free carriers under illumination [20]; (c) Photothermoelectric effect: The phenomenon in which a temperature gradient induced by photon absorption generates thermoelectric voltage in the material [21]; (d) Pyroelectric effect: The phenomenon where temperature changes under illumination induces charge generation on the surface of the material [22]; (e) Photogating effect: After light illumination generates charge carriers, some excess carriers become trapped, affecting the transport of charge carriers in the conductive channel. Furthermore, the extended lifetime of the trapped carriers influences the performance of the photodetector [23]. These effects work synergistically during the optoelectronic conversion process in photodetectors, collectively determining their overall performance. Therefore, enhancing the performance of photodetectors by leveraging these effects is both direct and effective. However, directly controlling these effects is extremely challenging. Most researchers adjust the materials or structures of photodetectors to indirectly modulate these effects, thereby improving their performance. For example, Claro et al. successfully controlled the photovoltaic effect in InAs quantum dot-based photodetectors by adjusting the type and density of quantum dots [24]. Compared to conventional InAs quantum dots, the photodetector with controlled quantum dots achieved a threefold increase in the specific detectivity at zero bias voltage. Zheng et al. enhanced the photothermoelectric effect to improve the infrared detection performance of photodetectors by modifying Ag nanoparticles on carbon nanotubes to form an Ag/carbon nanotube heterojunction [21]. The photodetector demonstrated excellent photothermal conversion capability and electrical energy conversion under infrared laser illumination, with an ultra-high temperature gradient and output voltage, showcasing outstanding optical detection performance. Additionally, some researchers have achieved control over one or more effects through specific material selection and structural design. For instance, Nan et al. used the ferroelectric material BaTiO<sub>3</sub> to fabricate a photodetector, successfully controlling both the photovoltaic and photothermoelectric effects [25]. Experimental results showed that the photodetector based on BaTiO<sub>3</sub> films significantly improved responsivity and specific detectivity under the photovoltaic-thermoelectric coupling effect, with an enhancement of over 260 %.

The photogating effect, as one of the key phenomena in the optoelectronic conversion process of photodetectors, can significantly enhance the performance of photodetectors when properly controlled during their design. Generally, the strategies for modulating the photogating effect can be broadly divided into two categories: One approach involves modifying the photodetector material to control the photogating effect. Due to the limited size in at least one dimension and lower screening effects in low-dimensional materials, defects and impurities often become charge carrier trapping centers rather than recombination centers, as is the case with conventional materials [23]. This leads to the observable presence of the photogating effect in photodetectors based on low-dimensional materials. As a result, many researchers have used low-dimensional materials to introduce the photogating effect in photodetectors, thus improving their performance [23,26–29]. However, the limited thickness of low-dimensional materials results in poor light absorption, which is highly detrimental to their application in photodetectors [30,31]. Therefore, an alternative and more common strategy for applying the photogating effect is to construct heterojunctions that introduce trap states. This approach is more versatile and can incorporate novel materials to further enhance photodetector performance, such as perovskite materials [32,33].

Based on the widespread applications of the photogating effect, this paper will review the principles behind the generation of the photogating effect, followed by an introduction to its impact on the performance of photodetectors. This includes effects on gain and bandwidth, responsivity and response time, specific detectivity, as well as transfer characteristic curves. Finally, this review not only summarizes the applications of the photogating effect in photodetectors based on low-dimensional materials, but also extends the discussion to perovskite-based devices. In contrast to previous reviews by Fang et al. [23] and Shin et al. [33], which primarily focus on conventional low-dimensional systems, this work highlights the integration of the photogating effect with emerging perovskite materials, providing a valuable reference for future research directions.

## 2. Principle of photogating effect

The photogating effect refers to the phenomenon where, after carrier

generation induced by light exposure, a portion of the excess carriers become trapped, thus affecting the carrier transport within the conductive channel (Fig. 1a illustrates the modulation of electron transport by hole traps in valence band tail states). This effect involves the drift of the Fermi level and conductivity modulation caused by the carrier trapping effect of photogenerated carriers, and holds significant application potential in novel photodetectors [23,26].

When semiconductor materials are illuminated, if the energy of the incident photons  $h\nu$  exceeds the material's bandgap  $E_g$ , electron-hole pairs can be generated between the conduction and valence band [34]. The generation rate of photogenerated carriers per unit volume is expressed as:

$$g = \frac{\eta P}{h\nu V} \quad (2-1)$$

where  $\eta$  is the quantum efficiency,  $P$  is the incident light power,  $h\nu$  is the energy of a single photon, and  $V$  is the volume of the material. In an ideal scenario, photogenerated carriers undergo recombination within the semiconductor, following the carrier dynamics equation:

$$\frac{dn}{dt} = g - \frac{n}{\tau} \quad (2-2)$$

where  $n$  is the instantaneous concentration of free photogenerated carriers and  $\tau$  is the carrier lifetime. Under steady-state conditions ( $dn/dt=0$ ), the free carrier concentration reaches an equilibrium value:

$$n_{\text{light}} = g\tau \quad (2-3)$$

However, photogenerated carriers not only participate in direct recombination but can also be trapped by defect states within the material. These trapped carriers may remain in the traps for extended periods, thereby influencing the material's electronic structure and optoelectronic response characteristics [23].

To describe the impact of the photogating effect on photodetectors, we introduce the concept of the Fermi level. The Fermi level  $E_f$  reflects the statistical distribution of electrons within a material, and its position is influenced by the carrier concentration. For non-degenerate semiconductors, the concentrations of electrons and holes can be described by the following equations:

$$n = N_c \exp\left(-\frac{E_c - E_f}{kT}\right) \quad (2-4)$$

$$p = N_v \exp\left(-\frac{E_f - E_v}{kT}\right) \quad (2-5)$$

where  $N_c$  and  $N_v$  are the effective density of states in the conduction and valence bands,  $E_c$  and  $E_v$  are the energy levels at the conduction band minimum and valence band maximum,  $k$  is the Boltzmann constant, and  $T$  is the temperature (The relative positions of  $E_c$ ,  $E_f$ , and  $E_v$  are shown in Fig. 1b). When photogenerated carriers are captured by trap states, it leads to an asymmetric change in the concentration of free carriers, which in turn affects the position of the Fermi level. The drift of the Fermi level  $\Delta E_f$  can be approximated as:

$$\Delta E_f = kT \ln \frac{n_{\text{light}}}{n_0} \quad (2-6)$$

where  $n_0$  is the concentration of free carriers in the dark state, and  $n_{\text{light}}$  is the concentration of free carriers under illumination. In different types of semiconductors, the direction of the Fermi level drift varies: (a) For n-type semiconductors: If photogenerated holes are captured by traps, the concentration of free electrons increases, causing the Fermi level to shift upward, closer to the conduction band. (b) For p-type semiconductors: If photogenerated electrons are captured by traps, the concentration of free holes increases, causing the Fermi level to shift downward, closer to the valence band [35–38]. The carrier trapping effect of defect states enables dynamic modulation of the Fermi level under illumination,

resulting in the photogating effect.

Obviously, trap states play a crucial role in the photogating effect. Generally speaking, there are three main causes of trap state formation in photodetectors: (a) Trap states induced by defects and impurities. These trap states are quite common in photodetectors based on low-dimensional materials. This is due to the extremely limited size of the material in one dimension and the relatively low screening effect, where impurities and defects within the material act as trap centers for carriers, capturing specific types of carriers. This is different from photodetectors based on thin films, where impurities and defects typically serve as recombination centers for carriers [23,26,28,33]. The type of photogenerated carriers captured by trap states is related to the Fermi level. In n-type channels, hole trap states are typically present, while in p-type channels, electron trap states are more common [28]. It can be observed that the traps in the channel usually capture minority carriers. If the majority of carriers are captured, it leads to a decrease in conductivity, which results in a negative photoconductivity (NPC) phenomenon [39]. (b) Charge exchange at the heterojunction. Specifically, at the heterojunction, due to the differences in the band structure and electron affinity of the two materials, charge transfer occurs when illuminated or under the influence of an external electric field. This charge exchange leads to the generation of a local electric field in the heterojunction region, which in turn forms deep-level trap states near the heterojunction interface. (c) Charge accumulation at the heterojunction. In photodetectors, when there is a significant difference in the conductivity of the two materials at the heterojunction, photogenerated carriers may accumulate at the interface, especially in the absence of sufficient charge transport channels or electric fields. This charge accumulation leads to an enhanced interface electric field, which induces local carrier piling. This phenomenon not only alters the electric field distribution near the interface but also may form a series of deep traps at the interface, trapping some carriers before recombination, ultimately affecting the photodetector's optoelectronic response. Both charge exchange and charge accumulation at the heterojunction result in the trapping of specific carriers, thereby generating the photogating effect [23]. It is worth mentioning that the trapping of carriers at the heterojunction can also give rise to the NPC phenomenon. In fact, the NPC effect is closely related to the photogating effect, as will be discussed in detail later.

Compared with the latter two cases, the trap states discussed in case (a) are not associated with heterojunction formation. As a result, structural strategies commonly employed in the latter—such as heterojunction engineering to modulate trap states—are difficult to apply, leading to relatively poor tunability. Nevertheless, trap states are not entirely uncontrollable defects; they can also be intentionally introduced and modulated through rational design and engineering strategies [40,41]. Common approaches include: (1) Doping and ion substitution, where the concentration and type of dopants can be adjusted to introduce trap states with specific energy levels and characteristics within the material [42]; (2) Plasma treatment, such as  $O_2$  plasma exposure, which has been widely used to create oxygen-related surface traps in two-dimensional materials. In addition, several advanced characterization techniques have been employed to quantitatively identify and correlate trap states with structural properties [43]. These include deep-level transient spectroscopy (DLTS), photothermal deflection spectroscopy (PDS), photoinduced current transient spectroscopy (PICTS), and time-resolved photoluminescence (TRPL), all of which enable precise defect profiling and modeling [40,41,43]. However, compared with structural optimization, these trap engineering approaches are generally more difficult to control. Moreover, heterostructure-based structural design can synergistically combine the advantages of different materials, resulting in superior photodetector performance. Therefore, heterojunction engineering remains the more commonly adopted and practical method for trap-state modulation in current device architectures.

### 3. Impact of photogating effect on photodetector performance

The impact of the photogating effect on photodetectors mainly manifests in the following aspects: (i) In photodetectors, the photogating effect can enhance the sensitivity by trapping carriers, thus prolonging their dwell time within the detector. This means that the detector can respond more strongly to weak light signals, improving its sensitivity. By properly designing the electric field or structure, the detector's sensitivity to low light intensities can be enhanced, enabling it to even respond to faint light signals [46]. (ii) By adjusting the degree of carrier "trapping", the photogating effect helps the detector extend its dynamic range between high and low light intensities. This effect allows the detector to handle a broader range of light intensity variations without distortion caused by overexposure [47,48]. (iii) In certain applications, the photogating effect can reduce interference from environmental noise. Because carriers are trapped at specific locations, the detector becomes less sensitive to rapidly changing noise signals, thus enhancing the signal-to-noise ratio [49]. (iv) In imaging systems such as CMOS sensors, the photogating effect can enhance the light capture capability of each pixel and improve image quality. In many dynamic scenarios, the photogating effect helps capture momentary changes in the image, preventing the response delay of traditional photosensitive materials to fast-moving objects [50]. (v) The photogating effect also finds applications in quantum-level detection, where it helps develop ultra-efficient quantum detectors by making more sensitive responses to different wavelengths of light sources. This effect holds great potential, especially in single-photon detection technologies [51].

In the research and application of photodetectors, several performance parameters determine their overall performance. To thoroughly investigate the impact of the photogating effect on these parameters, it is first necessary to clarify the core performance indicators of the detectors. These performance parameters not only relate to the fundamental characteristics of the photodetectors but also directly determine their performance in practical applications.

#### 3.1. Gain and bandwidth

The gain is defined as the ratio of the number of charge carriers collected per unit time to the number of photons absorbed, and it characterizes the ability of a photodetector to amplify optical signals. This parameter is one of the key factors to consider when applying photodetectors. The photogating effect has a significant impact on photodetectors by dramatically enhancing their gain, resulting in a substantial improvement in the detection of weak signals [52].

For photodetectors dominated by the photoconductive effect, their gain can be approximately estimated using the following equation:

$$G = \frac{\tau}{\tau_T} \quad (3-1)$$

In this equation,  $\tau$  represents the lifetime of minority carriers in the photodetector, and  $\tau_T$  denotes the transit time of majority carriers in the conductive channel [23]. As mentioned earlier, the photogating effect refers to the phenomenon where a portion of charge carriers are trapped, modulating the carrier transport in the conductive channel. It is evident that modulating carrier transport in the conductive channel leads to changes in the photodetector's conductivity. Therefore, the photogating effect can be considered a special case of the photoconductive effect, and the equation (3-1) can also be used to estimate the gain of photodetectors dominated by the photogating effect [23,53]. Specifically, taking an n-type semiconductor as an example, the majority carriers responsible for conduction are electrons, while the minority carriers are holes. In n-type channels, hole trap states often exist, capturing holes and delaying their recombination, which significantly extends the lifetime of minority carriers. In this case, the transit time of electrons in the conductive channel is  $\tau_T$ , and the lifetime of holes in the trap states is  $\tau$ . When  $\tau > \tau_T$ , the hole lifetime exceeds the electron transit time in the

conductive channel. When excess electrons reach the anode, an electron from the cathode enters the conductive channel to maintain charge neutrality. This process continues until the excess electrons recombine with holes. Since  $\tau > \tau_T$  in this scenario, the gain  $G$  of the photodetector is greater than 1. Thus, the existence of trap states in the photogating effect leads to an extension of carrier lifetime, effectively enhancing the gain of the photodetector. When other conditions remain unchanged, the longer the carriers remain in the trap states, the higher the gain of the photodetector.

Equation (3-1) is applicable to the calculation of the gain for all photodetectors dominated by the photoconductive effect. However, for photodetectors dominated by the photogating effect, the gain can be expressed in a different form. Since a type of photo-generated charge carrier is captured and trapped in trap states, these carriers can act as a local gate voltage ( $\Delta V_g$ ) to modulate the channel conductivity. Under optical illumination, the photo-generated current in the photodetector is given by:

$$I_{ph} = \frac{\partial I_d}{\partial V_g} \times \Delta V_g = g_m \Delta V_g \quad (3-2)$$

In this equation,  $g_m$  is the transconductance, and its sign depends on the polarity of the majority carriers in the conductive channel, while the sign of  $\Delta V_g$  depends on the type of local trap state [23]. If the majority carriers are holes and the trap states capture electrons, then  $g_m < 0$  and  $\Delta V_g < 0$ , resulting in a positive photocurrent. It is important to note that both  $g_m$  and  $\Delta V_g$  can be either positive or negative. According to equation (3-2), the value of  $I_{ph}$  can be negative, which corresponds to NPC [39]. According to equation (3-2), under optical illumination, the number of charge carriers collected by the photodetector per unit time is given by  $g_m \Delta V_g / e$ , where  $e$  is the unit charge. The number of absorbed photons is  $\eta \cdot PA / h\nu$ , where  $P$  is the incident light power density,  $A$  is the effective light absorption area of the photodetector, and  $h$  is Planck's constant. Therefore, based on the definition of gain, the ratio of the number of charge carriers collected per unit time to the number of absorbed photons is given by:

$$G = \frac{g_m \Delta V_g / e}{\eta \cdot PA / h\nu} \quad (3-3)$$

From equation (3-3), it can be observed that the gain decreases as  $P$  increases. This is partly because the trap states are gradually filled under the influence of light. Once all the trap states are completely filled at a certain light power intensity, stronger light power will excite free charge carriers that cannot be captured, leading to a decrease in the average carrier lifetime. As shown in equation (3-1), this ultimately results in a reduction in gain.

Bandwidth refers to the range of optical signal frequencies that a photodetector can respond to, measured in Hz [54]. It reflects the dynamic response capability of the detector, i.e., the frequency at which variations in the incident optical signal can still be accurately converted into an electrical signal by the detector. The larger the bandwidth, the stronger the detector's ability to respond to rapidly changing signals. However, in photodetectors, there is generally an inverse relationship between gain and bandwidth. The maximum bandwidth, unaffected by gain attenuation, is proportional to  $1/\tau$  [23], meaning that the longer the lifetime of minority carriers, the narrower the photodetector's bandwidth. At the same time, as shown in equation (3-2), gain is positively correlated with  $\tau$ . Therefore, in practical applications, it is necessary to strike a balance between bandwidth and gain to achieve optimal performance.

The gain-bandwidth product is typically used to measure the frequency range a photodetector can handle while providing high gain [55]. Specifically, it represents the product of the highest frequency the photodetector can process without distortion and the gain of the photodetector. When the gain-bandwidth product of a photodetector is fixed, a higher gain results in a smaller bandwidth. Therefore, to achieve



both high gain and wide bandwidth, it is necessary to improve the gain-bandwidth product. As a result, while utilizing the photogating effect to enhance the gain of photodetectors, researchers have also explored other methods to broaden the bandwidth and thereby increase the gain-bandwidth product, ultimately improving the photodetector's performance. Shen et al. employed the photogating effect to achieve a gain of up to  $1.9 \times 10^3$  in a photodetector based on two-dimensional tellurium nanoplates [56]. Simultaneously, they leveraged the photoconductive effect to achieve a relatively large bandwidth of 37 MHz at the 1.55  $\mu\text{m}$  communication wavelength, resulting in an improved gain-bandwidth product and significantly enhancing the photodetector's performance.

### 3.2. Responsivity and response time

Responsivity is a measure of a photodetector's ability to convert incident optical signals into electrical signals. It is defined as the ratio of photocurrent to incident optical power, expressed by the equation  $R = I_{ph}/P_{in}$ , with the unit of  $\text{A} \cdot \text{W}^{-1}$  [57]. Responsivity reflects the detector's efficiency in converting optical signals at a specific wavelength; a higher responsivity indicates better conversion efficiency for signals at that wavelength. The presence of the photogating effect leads to an enhancement of the photocurrent, thereby improving the responsivity of the photodetector and enhancing its efficiency in converting optical signals at a particular wavelength.

Response time characterizes the speed at which a detector responds to variations in optical signals, encompassing both rise time ( $\tau_r$ ) and decay time ( $\tau_d$ ). The rise time ( $\tau_r$ ) is the time it takes for the current to reach a stable level after receiving photon flux, while the decay time ( $\tau_d$ ) is the time required for the current to stabilize after the photon flux is removed [23]. Generally, rise time is defined as the time it takes for the net photocurrent to increase from 10 % to 90 %, while decay time is defined as the time it takes for the net photocurrent to decrease from 90 % to 10 % [26]. A shorter response time indicates that the detector can more rapidly follow changes in the optical signal, which is especially important for high-speed optoelectronic detection applications. In the case of the photogating effect, the trapping states significantly prolong the release process of captured charge carriers, causing them to remain in the trapped state for a long time after the light is turned off. This makes it difficult for the trapped carriers to detrapped and recombine. For instance, Guo et al. fabricated a mid-infrared black phosphorus photodetector with shallow trap states [58]. Despite the shallow nature of the traps, the lifetime of the captured charge carriers in this photodetector still reached 0.13 ms. Such a long carrier lifetime greatly increases the difficulty of carrier recombination after light cutoff, leading to a significant slowing of the response time.

Thus, the presence of the photogating effect creates a trade-off between the responsivity and response time of photodetectors, as evidenced by the relationship between responsivity and response time for common photodetectors shown in Fig. 1c [44]. Typically, increasing the responsivity of a detector requires a higher number of generated charge carriers. In this regard, the photogating effect is advantageous, as the charge carriers trapped in trap states delay recombination, resulting in a higher responsivity for photodetectors exhibiting the photogating effect. However, the presence of trap states also significantly prolongs the lifetime of the trapped charge carriers, which slows down the response speed of the photodetector. Consequently, higher responsivity is often accompanied by longer response time. Therefore, the selection of photodetectors should be made based on a trade-off tailored to the specific application. For example, in optical communication, where extremely short response times are required to track rapidly changing optical signals, designs with lower responsivity are typically chosen. In contrast, certain applications, such as in astronomy, require higher responsivity to detect faint optical signals, where response time becomes less critical.

Although there is a classical trade-off relationship between responsivity and response time, researchers have proposed various strategies in

recent years to overcome this inherent limitation. Fu et al. proposed and experimentally demonstrated a novel "photo-driven" semimetal-semiconductor field-effect transistor (photo-sMESFET) based on a graphene/Si-on-insulator heterostructure, achieving simultaneous optimization of responsivity, response speed, and dark current [59]. Furthermore, Fu et al. also developed a polarity-tunable field-effect phototransistor with a graphene/ultrathin  $\text{Al}_2\text{O}_3$ /silicon structure [60]. This device can modulate the gate control effect under illumination, enabling the transfer characteristics to switch from unipolar to bipolar and significantly enhancing the photocurrent signal. The introduced ultrathin  $\text{Al}_2\text{O}_3$  interlayer not only adjusts the band structure but also achieves a synergistic improvement in device performance. The device exhibits a high responsivity exceeding  $10^5 \text{ A} \cdot \text{W}^{-1}$ , a 3 dB bandwidth of 100 kHz, and a gain-bandwidth product of  $9.14 \times 10^{10} \text{ s}^{-1}$ , with a specific detectivity as high as  $1.91 \times 10^{13}$  Jones. This study demonstrates that through structural design and polarity modulation, the traditional trade-off between "high gain and slow response" in field-effect phototransistors can be effectively broken, providing a feasible approach for realizing next-generation photodetectors with high sensitivity and fast response.

### 3.3. Specific detectivity

The specific detectivity ( $D^*$ ) is used to describe the sensitivity of the detector to signals under a given noise environment. It reflects the detector's ability to detect signals at low light intensities, especially under conditions with high noise, indicating the effectiveness of the detector. The unit is Jones, and the equation is as follows:

$$D^* = \frac{\sqrt{A \cdot BW}}{NEP} \quad (3-4)$$

where  $A$  is the effective light absorption area of the photodetector,  $BW$  is the bandwidth, and  $NEP$  is the noise equivalent power [61]. Under the condition of constant  $A$ , the introduction of the photogating effect limits the bandwidth; however, its effect on reducing the  $NEP$  is quite significant. This is because the photogating effect effectively regulates the carrier recombination process, reducing the loss of useless carriers and thereby improving the detector's response efficiency to optical signals [61,62]. The photogating effect works by trapping carriers and temporarily delaying their recombination process, which reduces the opportunities for carriers to recombine during unnecessary time intervals, resulting in a stronger photocurrent under the same conditions. Specifically, when light illuminates the detector, the photo-generated carriers are temporarily trapped in the trap states and are later released under appropriate conditions to participate in the current transport through the conductive channel. This process enhances the detector's response to weak light signals, enabling it to maintain a high  $D^*$  even under low-light conditions. By applying pulse voltages to adjust the depth of the trap states and the capture/release rates of carriers, the photogating effect can optimize the dynamic behavior of carriers, thus improving the photodetector's photoelectric conversion efficiency and  $D^*$ . Additionally, the photogating effect adjusts the detector's response to different light intensities, ensuring efficient detection performance under both high and low light conditions. This is crucial for enhancing the dynamic range of the detector.

### 3.4. Transfer characteristics curve and NPC

The transfer characteristics curve of a photodetector is one of the key methods for evaluating its performance. It shows the variation in conductivity or current of the detector under different gate voltages. The photogating effect, as a technique to enhance the performance of photodetectors, involves modulating the carrier transport in the conductive channel by charge trap states under illumination. This modulation can significantly affect the transfer characteristics curve. Specifically, if the

**Table 1**

The comparison between PPC and NPC [63].

|                     | PPC                                                                                               | NPC                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Definition          | Current increases after illumination                                                              | Current decreases after illumination                                                                |
| Cause               | Separation of photogenerated electron-hole pairs; enhanced conduction in conduction/valence bands | Majority carriers are trapped, or photogenerated carriers are opposite in type to majority carriers |
| Role of Trap States | Trap minority carriers, prolong lifetime, increase gain                                           | Trap majority carriers, reduce effective carrier density in conduction channel                      |
| Transfer Curve      | Transfer curve shifts left with increased current after illumination (e.g., point A in Fig. 1c)   | Transfer curve shifts right with decreased current after illumination (e.g., point B in Fig. 1c)    |
| Behavior            |                                                                                                   |                                                                                                     |
| Schematic           |                                                                                                   |                                                                                                     |
| Diagram             | <p>Graphene-Ge junction forward biased</p>                                                        | <p>Graphene-Ge junction reverse biased</p>                                                          |

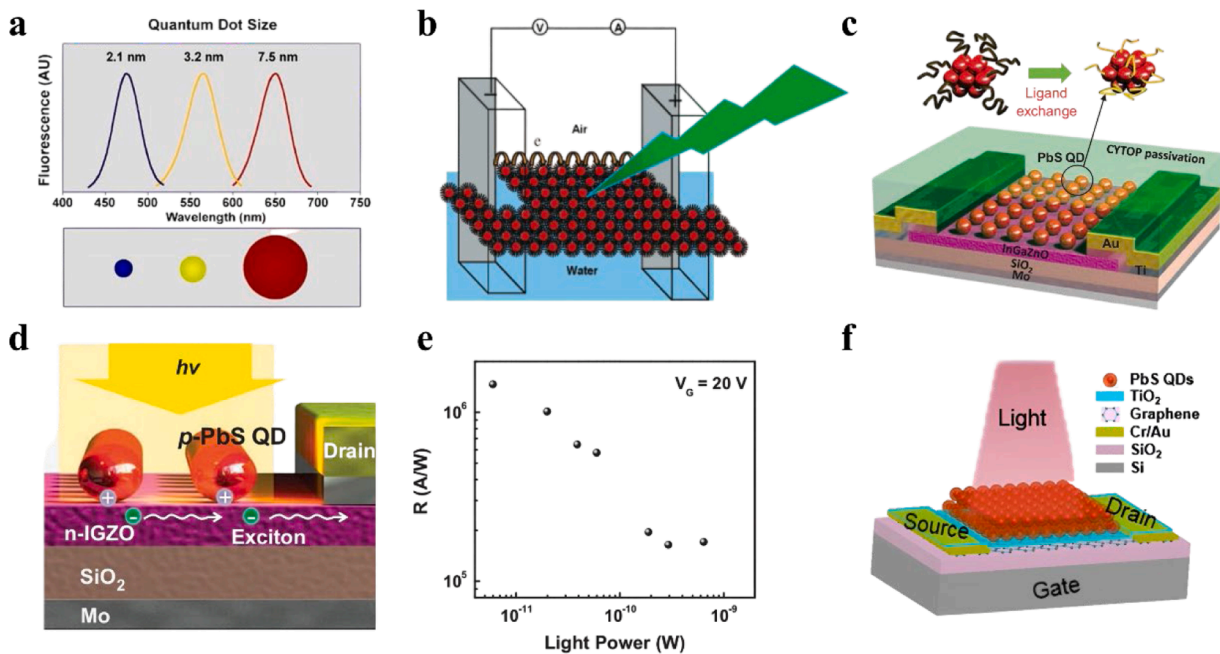
trapped carriers in the trap states are holes, they will carry a positive charge, resulting in a positive  $\Delta V_g$  and a leftward shift of the transfer characteristics curve. Conversely, if the trapped carriers are electrons, they will carry a negative charge, causing a negative  $\Delta V_g$  and a rightward shift of the transfer characteristics curve, as shown in Fig. 1d (the black curve represents dark current, while the red curve shows the shift in the presence of photogating effect under illumination) [26]. Furthermore, since the magnitude of  $\Delta V_g$  is related to the gate voltage, the actual shift is not always uniform; typically, there is a slight vertical shift accompanying the horizontal shift, as shown by the blue curve in Fig. 1d [23]. Therefore, the photocurrent generated by a photodetector exhibiting the photogating effect may be greater than the dark current (Point A in Fig. 1d) or smaller than the dark current (Point B in Fig. 1d), corresponding to positive photoconductivity (PPC) and negative photoconductivity (NPC) phenomena.

PPC is relatively common in photodetectors, whereas the occurrence of NPC is often associated with the photogating effect. Specifically, there are two main causes for the generation of NPC: the first is when the type of photo-generated carriers injected into the channel is opposite to that of the majority carriers in the channel, and the second is when the type

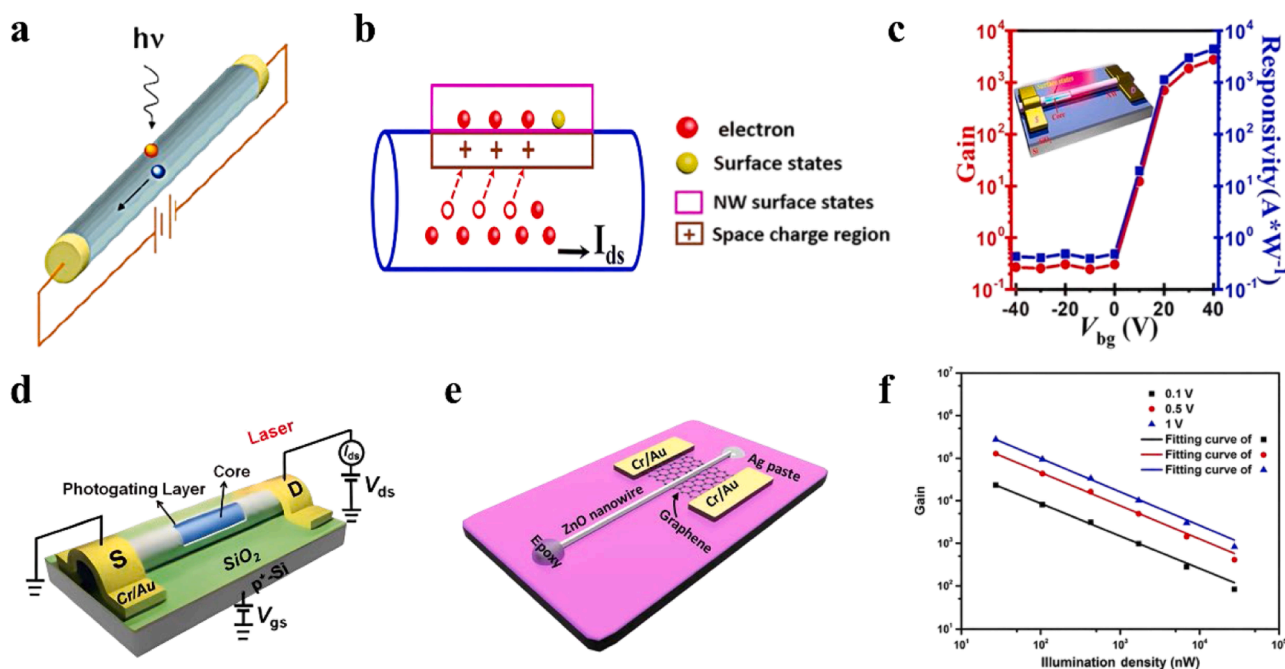
of photo-generated carriers trapped in the trap states is the same as the majority carriers in the channel [39]. The reason described in the latter is shown in Fig. 1e [45]. This differs significantly from PPC (Table 1) [63]. The former is rarely observed in typical photodetectors, while the latter is closely linked to the presence of trap states, and thus, is strongly connected to the photogating effect.

As mentioned earlier, there are three main causes for the formation of trap states: impurities or defects, charge exchange at heterojunctions, and charge accumulation. Considering the mechanisms behind NPC, for NPC to occur, the trap states must be multi-carrier trap states, which are difficult to form in single-material systems. This is because multi-carrier trap states are usually already filled before illumination, and thus NPC is more likely to occur in heterostructures [28,64]. Zhu et al. studied the interface dynamics of photo-generated carriers in  $\text{Cs}_3\text{Bi}_2\text{I}_9/\text{Au}$  heterostructure nanocomposites and found that the introduction of Au nanoparticles led to a transition from PPC to NPC [65]. They attributed this transition to the capture and recombination processes of photo-generated carriers at the heterointerface, which is precisely the mechanism behind the photogating effect.

NPC contributes to expanding the functionality of traditional



**Fig. 2.** (a) Relationship between the emission wavelength and size of CdSe quantum dots [79]; (b) CdSe QDs at the air-water interface [80]; (c) Schematic structure of PbS QDs/InGaZnO heterostructure optoelectronic detector [81]; (d) Schematic of the photogating effect in the PbS QDs/InGaZnO heterostructure optoelectronic detector [81]; (e) Responsivity of the PbS QDs/InGaZnO heterostructure optoelectronic detector under different optical excitation powers [81]; (f) Schematic structure of the graphene/TiO<sub>2</sub> film/PbS QDs heterostructure optoelectronic detector [85].



**Fig. 3.** (a) Schematic illustration of the working principle of NW-based optoelectronic detectors [86]; (b) Schematic representation of carrier trapping by surface states in an InAs NW-based optoelectronic detector [88]; (c) Gain and responsivity of the InAs NW-based optoelectronic detector under different back gate voltages [88]; (d) Schematic structure of the InAs NW-based optoelectronic detector fabricated via chemical vapor deposition [91]; (e) Schematic illustration of the ZnO NWs/graphene heterojunction optoelectronic detector [93]; (f) Gain of the ZnO NWs/graphene heterojunction optoelectronic detector under varying optical excitation power [93].

photodetectors and holds promise for application in low-power, high-speed photodetectors with fast frequency response [66,67]. Moreover, when combined with PPC, NPC can be designed into photonic logic gates [68]. The presence of NPC helps to enhance the stability of devices, particularly under long-term operation or high light intensity conditions. NPC works by limiting the excessive migration or accumulation of photo-generated carriers, effectively suppressing the over-diffusion of carriers within the material. This prevents an excessive number of carriers from being trapped or recombining, thereby reducing performance fluctuations caused by an excess of carriers. This stability is crucial in applications that require long-duration operation and high environmental reliability. Additionally, NPC allows for the modulation of photoconductivity by adjusting external light intensity or material properties. This tunability enables certain photodetectors to dynamically adjust their responsivity based on changes in the operating environment, especially under complex and fluctuating lighting conditions. With appropriate regulation, photodetectors can maintain optimized response performance across a range of light intensities [39,64].

#### 4. Photogating effect in low dimensional photodetectors

Low-dimensional materials refer to those with at least one dimension at the nanoscale, which can be classified into zero-dimensional (0D), one-dimensional (1D), and two-dimensional (2D) materials. Quantum dots (QDs) are considered zero-dimensional materials, while nanowires (NWs) are one-dimensional. In recent years, graphene, a representative 2D material, has gained significant attention. Additionally, other 2D materials such as MoS<sub>2</sub> and WSe<sub>2</sub> have also found extensive applications [69,70]. Due to their nanoscale dimensions and excellent optoelectronic properties, low-dimensional materials exhibit outstanding performance in optoelectronic detectors. As a result, optoelectronic detectors based on low-dimensional materials have become a focal point of research in recent years [71–76]. Thanks to the presence of the photogating effect, optoelectronic detectors based on low-dimensional materials have achieved remarkable performance, particularly with the significant

enhancement in gain attributed to the photogating effect.

##### 4.1. Quantum dots

Quantum dots (QDs), often referred to as “artificial atoms”, exhibit unique optoelectronic properties due to their quantum-sized dimensions in all three spatial directions [77,78]. By controlling the size of QDs during synthesis, their emission wavelength can be tuned across a wide spectral range. Fig. 2a shows the relationship between the emission wavelength of CdSe QDs and their size [79]. Moreover, the surface states of QDs can significantly impact their optoelectronic performance. Pradhan et al. investigated the conductivity of CdSe QDs at the air-water interface (Fig. 2b) and confirmed the surface trapping effect on photo-excited electrons based on the voltammetric characteristics of the QDs [80]. In other words, the surface of these QDs possesses electron trap states, which can induce the photogating effect. In fact, most QDs exhibit some degree of charge carrier trapping due to their surface states. However, optoelectronic detectors based on individual QDs suffer from low carrier mobility, which severely limits their performance. Consequently, QDs are often combined with other materials to enhance their performance. By utilizing the formation of heterostructures, the photogating effect can be introduced within the optoelectronic detector, further improving its performance [23,77]. Hwang et al. integrated PbS QDs with InGaZnO (Fig. 2c), and the hole trapping effect at the PbS QDs/InGaZnO heterojunction induced the photogating effect [81]. As shown in Fig. 2d, the trapped holes modulated the electron transport in InGaZnO, enhancing the optoelectronic performance of the detector. Through the regulation of the photogating effect, the optoelectronic detector achieved a responsivity of  $>10^6 A \cdot W^{-1}$  under low light excitation power (Fig. 2e) and a  $D^*$  of  $10^{13}$  Jones at a wavelength of 1000 nm. In addition to forming heterojunctions with conventional metal oxides, QDs can also be combined with 2D materials to construct heterojunction-based optoelectronic detectors. QDs can compensate for the poor light absorption ability of 2D materials, while the 2D materials, serving as conductive channels, help overcome the low carrier mobility



of the QDs. Konstantatos et al. introduced PbS QDs onto the typical 2D material graphene to construct a heterostructure [82]. In the QDs layer, strong and tunable light absorption generates charge carriers, which are transferred to graphene. Due to the high charge mobility in graphene and the photogating effect resulting from the trapping of charges at the QDs layer and heterojunction, the optoelectronic detector based on this structure achieved exceptional performance: a gain of  $10^8$ , responsivity of  $10^7 \text{ A}\cdot\text{W}^{-1}$ , and a  $D^*$  of  $7 \times 10^{13}$  Jones. Similarly, Ahn et al. fabricated a PbS QDs/graphene structure, achieving a responsivity of  $10^8 \text{ A}\cdot\text{W}^{-1}$  [83]. Beyond graphene, QDs are frequently combined with other 2D materials to obtain superior optoelectronic performance. Lian et al. developed a Si QDs/2D  $\text{MoS}_2$  heterostructure-based optoelectronic detector, which effectively utilized the advantages of both QDs and 2D materials [84]. The photogating effect was formed through charge accumulation at the Si QDs/ $\text{MoS}_2$  heterojunction, resulting in high detectivity and responsivity, particularly in the ultraviolet (UV) spectral range. Compared to monolayer  $\text{MoS}_2$ , the Si QDs/monolayer  $\text{MoS}_2$  heterojunction detector demonstrated a 100-fold increase in  $D^*$  at 365 nm (from  $1.0 \times 10^{12}$  to  $1.0 \times 10^{14}$  Jones) and an 89-fold increase in responsivity (from 66.7 to  $6.0 \times 10^3 \text{ A}\cdot\text{W}^{-1}$ ).

Although the photogating effect enables optoelectronic detectors to achieve remarkably high gain and responsivity, the trapping of charge carriers by trap states leads to delayed recombination, significantly limiting the response time. To address this issue, Ge et al. fabricated an optoelectronic detector based on a graphene/ $\text{TiO}_2$  film/PbS QDs heterostructure (Fig. 2f) [85]. By introducing the  $\text{TiO}_2$  film, the potential barrier within the QDs was reduced, effectively bridging the gap between high sensitivity and fast response time in visible-light detection. The presence of the photogating effect was confirmed through the shift in the transfer characteristic curve. As a result, the device demonstrated outstanding optoelectronic performance: under 635 nm illumination, it achieved a high responsivity of  $1.2 \times 10^4 \text{ A}\cdot\text{W}^{-1}$  and a high  $D^*$  of  $1.5 \times 10^{12}$  Jones, while maintaining a rapid decay time of 35 ms.

#### 4.2. Nanowires

Nanowires (NWs), with their exceptionally high surface-to-volume ratio and Debye length comparable to their small dimensions, exhibit excellent photosensitivity, making them a promising new material for optoelectronic detectors. The working principle of NW-based optoelectronic detectors is illustrated in Fig. 3a [86]. Due to the extremely high surface-to-volume ratio of NWs, their surface states have a significant impact on the optoelectronic performance of NW-based detectors. Under light illumination, surface depletion in NWs causes band bending at the surface, generating a strong electric field that facilitates the separation of photogenerated electron-hole pairs. This leads to the selective trapping of one type of charge carrier by the surface states, which then act as trap states, thereby inducing the photogating effect in the optoelectronic detector [87]. Zhang et al. fabricated an optoelectronic detector based on InAs NWs, where the surface of InAs NWs exhibits additional electron confinement energy levels [88]. Under illumination, surface states can capture electrons from the NW core, forming surface negative charge centers that modulate carrier transport within the core (Fig. 3b). By tuning the surface states through gate voltage and surface passivation, the device achieved a gain exceeding  $10^3$  and a high responsivity (Fig. 3c). Similarly, Liu et al. developed an InAs NW-based optoelectronic detector that, through the regulation of the photogating effect, achieved a high responsivity of  $4.4 \times 10^3 \text{ A}\cdot\text{W}^{-1}$  and a  $D^*$  of  $2.6 \times 10^{11}$  Jones [89]. Additionally, Hu et al. synthesized quasi-one-dimensional BiSeI microwires by precisely controlling material dimensions, obtaining optoelectronic properties similar to those of NWs due to their comparable size [90]. Under 515 nm illumination, the photogating effect induced by surface states enabled the detector to achieve an ultrahigh responsivity of  $7 \times 10^4 \text{ A}\cdot\text{W}^{-1}$  and a  $D^*$  of  $2.5 \times 10^{14}$  Jones.

However, although surface states can be modulated through various strategies to achieve optimal performance, they remain inherently

difficult to control in NWs. To circumvent the issue of uncontrollable surface states, Guo et al. employed chemical vapor deposition (CVD) to precisely regulate the growth of InAs NWs, thereby forming a self-assembled grating layer on the surface (Fig. 3d) [91]. Under CVD control, the InAs NWs developed a core/shell structure with a nearly perfect single-crystalline core and a defect-rich shell. Upon illumination, the defect-rich shell acts as a grating layer, trapping electrons generated in the core and creating an internal electric field that modulates the conductivity of the single-crystalline core. This mechanism mirrors the effect of surface states but offers significantly enhanced controllability. Owing to the regulation of surface states through the self-assembled grating layer, the optoelectronic detector achieved a high gain of approximately  $10^5$  at room temperature while maintaining a fast response time of 12 ms.

To further enhance the controllability of the photogating effect, constructing heterostructures to induce photogating has become a widely adopted strategy. Hojun et al. introduced p-type HgTe nanoparticles onto the surface of n-type ZnO NWs [92]. Under illumination, photogenerated electrons were trapped at the ZnO NW/HgTe nanoparticle heterojunction, thereby inducing the photogating effect. In this structure, the authors controlled the distribution of HgTe nanoparticles to regulate the photogating effect, leading to improved optoelectronic performance of the detector. Liu et al. fabricated a heterojunction optoelectronic detector by integrating ZnO NWs with the two-dimensional material graphene (Fig. 3e) [93]. By modulating the strain in ZnO NWs, photogenerated electrons were injected into graphene, while holes remained in ZnO NWs or at the heterojunction, effectively tuning the photogating effect. This device leveraged the ultrafast electron transport in graphene and the exceptionally long hole lifetime in ZnO NWs, achieving an ultrahigh responsivity of  $1.87 \times 10^5 \text{ A}\cdot\text{W}^{-1}$  and exceptionally high gain under varying optical excitation power ( $>10^4$ ) (Fig. 3f).

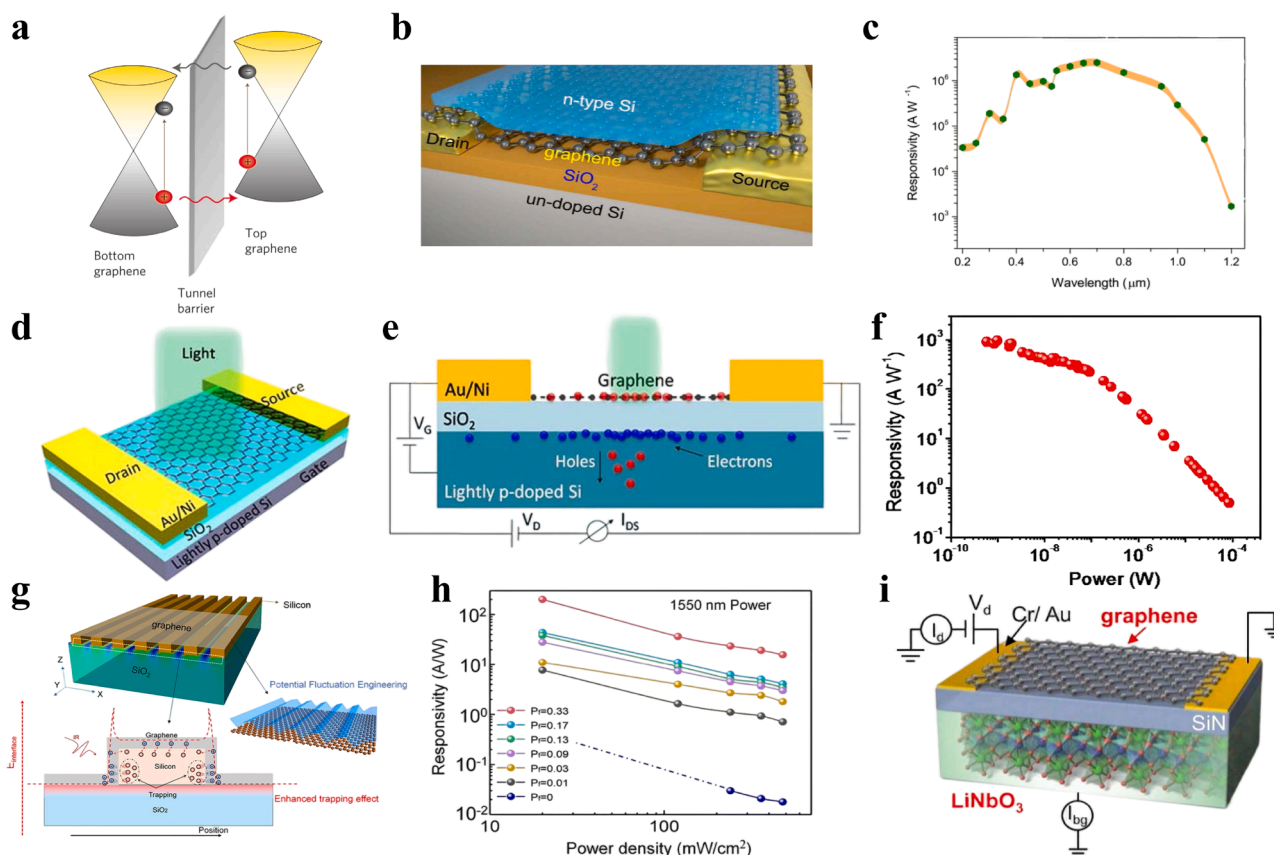
#### 4.3. Two-Dimensional materials

Two-dimensional (2D) materials refer to those that confine electron motion within a two-dimensional plane (1–100 nm scale). Since the advent of graphene in 2004, research on these materials has gradually become a hot topic in academia [94]. Graphene, due to its low defect density, ease of large-scale transfer, low cost, high surface area, and excellent carrier mobility [95,96], has demonstrated significant potential for use in high-frequency optoelectronic detectors [97–99]. However, due to its zero bandgap structure, graphene suffers from a short photogenerated carrier lifetime, which limits its application in optoelectronics. In addition to graphene, other 2D materials that are currently widely researched include transition metal dichalcogenides (TMDs) [100], black phosphorus (BP) [101] and hexagonal boron nitride (h-BN) [102]. These materials, with their rich electronic band structures, encompass a variety of types such as conductors, semiconductors, insulators, semimetals, and superconductors. Consequently, they exhibit broad spectral coverage, ranging from infrared to visible light, and even extending into the ultraviolet region [103,104]. The defects and impurities in 2D materials, due to their limited dimensions and relatively low screening effects, are prone to becoming carrier trap centers, thereby inducing the photogating effect. Moreover, the construction of heterostructures is another commonly used method for enhancing the performance of 2D material-based optoelectronic detectors by leveraging the photogating effect [23]. Besides using individual two-dimensional materials, constructing heterojunctions is also a common approach to enhance the performance of 2D material-based photodetectors via the photogating effect [23].

##### 4.3.1. Graphene

Since the discovery of graphene, it has attracted widespread attention due to its exceptional properties. As a representative of 2D materials, graphene-based photodetectors have demonstrated excellent performance, with vast potential applications [96]. In graphene-based





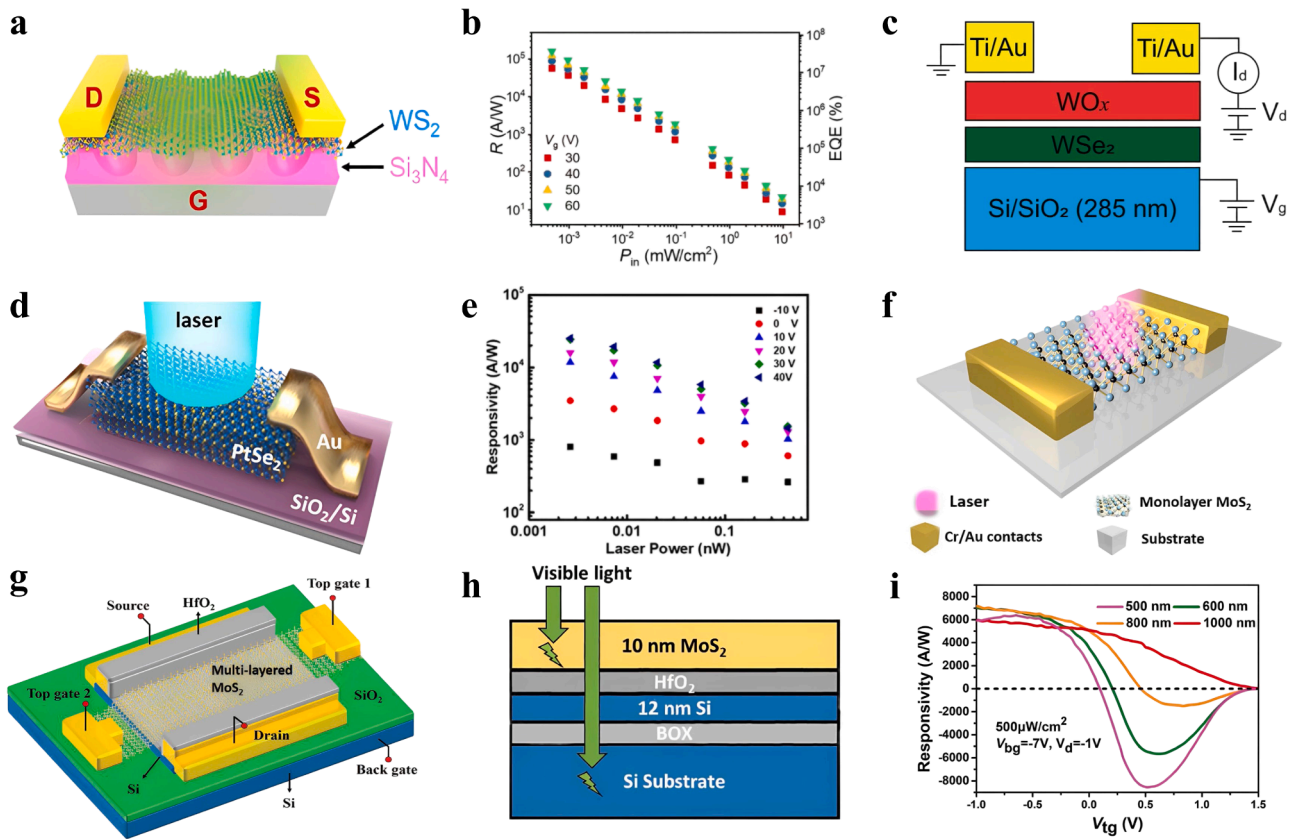
**Fig. 4.** (a) Schematic diagram of carrier transport in a photodetector based on two stacked monolayers of graphene [97]; (b) Schematic diagram of the graphene/Si heterostructure photodetector [105]; (c) Responsivity of the graphene/Si heterostructure photodetector at different wavelengths [105]; (d) Schematic diagram of the graphene/SiO<sub>2</sub>/lightly doped Si heterostructure photodetector [109]; (e) Schematic diagram of carrier distribution in the graphene/SiO<sub>2</sub>/lightly doped Si heterostructure photodetector under illumination [109]; (f) Responsivity of the graphene/SiO<sub>2</sub>/lightly doped Si heterostructure photodetector at different optical excitation powers [109]; (g) Schematic diagram of a photodetector with periodic nanoscale Si grating beneath graphene to construct potential fluctuation engineering [110]; (h) Responsivity of the photodetector with periodic nanoscale Si grating beneath graphene for different fluctuation periods and incident light power densities [110]; (i) Schematic diagram of the graphene/SiN/LiNbO<sub>3</sub> heterostructure photodetector [111].

photodetectors, graphene is often used as a channel material to form heterostructures with other materials. This is because the carrier transport speed in graphene is high, leading to a short carrier transit time in the channel. According to equation (3–1), the reduction in transit time results in an increase in gain [23,26]. Many researchers have utilized graphene as the channel material while designing specific structures to exploit the photogating effect, thus achieving high-performance photodetectors. For example, Liu et al. constructed a phototransistor with two stacked monolayers of graphene (with the top layer as the gate material and the bottom layer as the channel material), inserting a thin tunneling barrier between them. Under illumination, photogenerated electrons tunnel from the top graphene layer to the bottom graphene layer and transport within the bottom layer, while holes accumulate in the top layer, inducing a photogating effect on the bottom layer [97]. The schematic is shown in Fig. 4a. Under low excitation power, the device exhibits an ultra-high responsivity of over  $10^3 \text{ A} \cdot \text{W}^{-1}$ .

Although graphene exhibits excellent optoelectronic properties, its poor light absorption capability due to its ultra-thin thickness limits the further enhancement of graphene-based photodetector performance. Therefore, Wang et al. used electron beam evaporation to deposit a 10 nm thick n-type Si layer on the surface of graphene as the light absorption layer, as shown in the structural schematic in Fig. 4b [105]. Due to the energy level difference, a built-in electric field is generated at the graphene/Si interface. Under illumination, photogenerated electron-hole pairs are created in Si and separated at the graphene/Si interface under the built-in electric field. Holes are transferred to the graphene channel, while electrons are confined in the Si absorption

layer, generating a photogating effect. Under the influence of the photogating effect, this heterojunction photodetector exhibits extremely high light responsivity over a wide wavelength range ( $>10^6 \text{ A} \cdot \text{W}^{-1}$  in the range of 0.6–0.8  $\mu\text{m}$ ) (Fig. 4c). Similarly, Kang et al. designed a graphene-ReS<sub>2</sub> heterojunction photodetector using the stacking method, taking advantage of the direct bandgap and high light absorption of multilayer ReS<sub>2</sub> [106]. Due to the photonic effect induced by ReS<sub>2</sub> as the light absorption layer, the device achieves a maximum light responsivity of  $7 \times 10^5 \text{ A} \cdot \text{W}^{-1}$  under 400 nm illumination. Liu et al. constructed a graphene/BP heterostructure photodetector using the high light absorption rate of multilayer BP [107]. In this device, the top graphene layer serves a dual function as both the encapsulation and channel layers. The photodetector exhibits an ultra-high light responsivity of  $3.3 \times 10^3 \text{ A} \cdot \text{W}^{-1}$  and a gain as high as  $1.13 \times 10^9$  at a wavelength of 1550 nm. Yu et al. developed a MoTe<sub>2</sub>/graphene photodetector, in which multilayer MoTe<sub>2</sub> serves as the light absorption layer and monolayer graphene acts as the channel layer [108]. This structure effectively utilizes the strong light absorption and narrow bandgap of multilayer MoTe<sub>2</sub>, as well as the high carrier mobility and low resistance of graphene. Additionally, the photogating effect induced by electrons trapped in the MoTe<sub>2</sub> trap states significantly enhances the performance of the heterostructure photodetector compared to pure MoTe<sub>2</sub> and graphene. Notably, at 1064 nm, the device exhibits a high responsivity of  $970.82 \text{ A} \cdot \text{W}^{-1}$ , an exceptionally high gain of  $4.69 \times 10^8$ , and a  $D^*$  of up to  $1.55 \times 10^{11}$  Jones.

However, due to carrier recombination delayed by trap states, achieving ultra-high responsivity often comes at the cost of a slower



**Fig. 5.** (a) Schematic of the WS<sub>2</sub> optoelectronic detector with a Si<sub>3</sub>N<sub>4</sub> substrate [116]; (b) Responsivity and EQE of the WS<sub>2</sub> optoelectronic detector on a Si<sub>3</sub>N<sub>4</sub> substrate under different gate voltages and incident light power densities [116]; (c) Schematic of the WSe<sub>2</sub> optoelectronic detector with a self-limiting oxide layer, WO<sub>x</sub> [118]; (d) Schematic of the PtSe<sub>2</sub> optoelectronic detector [119]; (e) Responsivity of the PtSe<sub>2</sub> optoelectronic detector under different gate voltages and laser powers [119]; (f) Schematic of the MoS<sub>2</sub> optoelectronic detector [120]; (g) Schematic of the MoS<sub>2</sub>/HfO<sub>2</sub>/Si heterostructure optoelectronic detector [121]; (h) Schematic of the carrier distribution in the MoS<sub>2</sub>/HfO<sub>2</sub>/Si heterostructure optoelectronic detector under light illumination [121]; (i) Responsivity of the MoS<sub>2</sub>/HfO<sub>2</sub>/Si heterostructure optoelectronic detector under different gate-source voltages [121].

response time. To address the issue of slow response time caused by the photogating effect, Guo et al. developed a graphene/SiO<sub>2</sub>/lightly doped Si structure photodetector (Fig. 4d), using an interface gate control mechanism to balance the trade-off between ultrafast response and ultra-high responsivity [109]. Under illumination, photogenerated electron-hole pairs in the lightly doped Si are separated, with electrons accumulating at the SiO<sub>2</sub>/Si interface, forming a negative gate voltage that modulates the graphene conductive channel (Fig. 4e). This photodetector achieves a responsivity of  $>10^3$  A·W<sup>-1</sup> under low excitation power (Fig. 4f), while reducing the response time to 400 ns, and exhibiting a spectral response range extending from visible to near-infrared light.

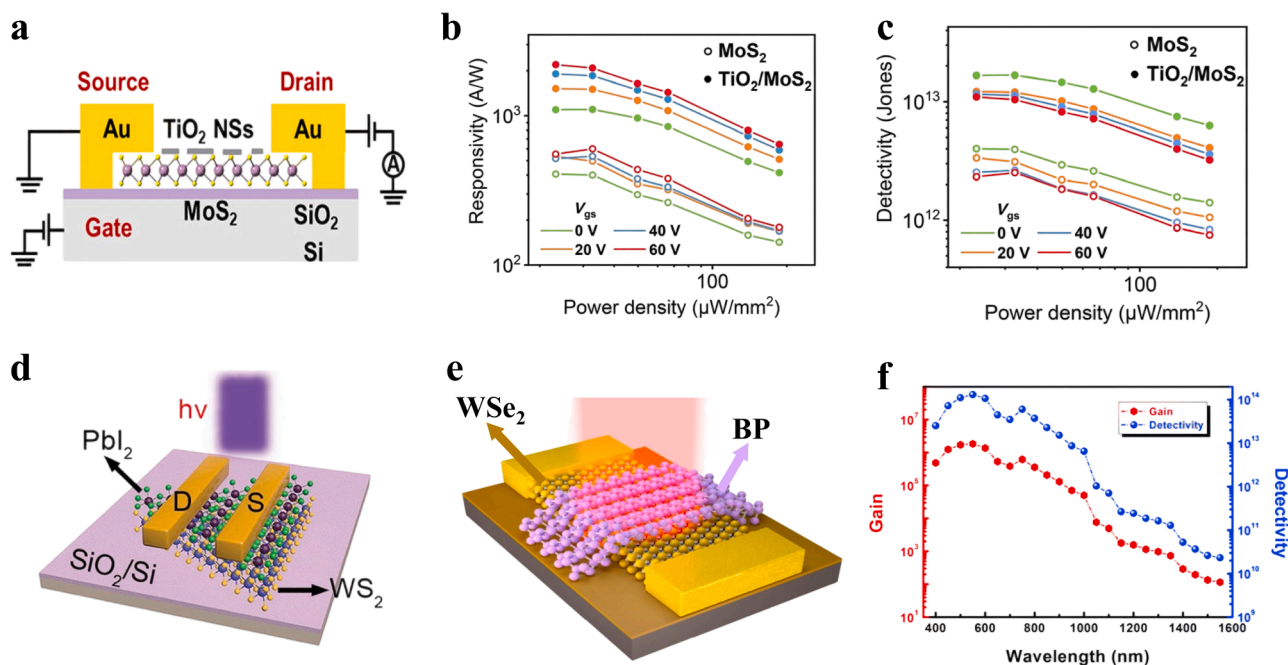
In addition to the mechanical stacking of two or more materials mentioned above, some researchers have proposed new approaches to utilize the photogating effect in graphene photodetectors. Jiang et al. introduced a periodic nanoscale Si grating structure beneath graphene, creating a potential fluctuation engineering effect that serves as an enhanced built-in potential, which separates and captures photogenerated carriers in graphene and transfers them to Si, thereby generating a photogating effect that modulates the conductivity of graphene [110]. The schematic is shown in Fig. 4g. Compared to devices without the periodic nanoscale Si substrate, the responsivity of devices with an optimized fluctuation period was increased by a factor of 100, reaching 240 A·W<sup>-1</sup> (Fig. 4h), with a  $D^*$  of up to  $3.4 \times 10^{12}$  Jones at a wavelength of 1550 nm and room temperature. Shimatani et al. used LiNbO<sub>3</sub> as the substrate and inserted a SiN layer between LiNbO<sub>3</sub> and graphene, as shown in Fig. 4i [111]. LiNbO<sub>3</sub> is a pyroelectric material capable of producing significant spontaneous polarization at room

temperature. This spontaneous polarization can cause carriers to enter a state similar to that of bound states, inducing a photogating effect. Additionally, SiN can absorb long-wavelength infrared radiation, increasing the temperature and enhancing both the pyroelectric effect and the photogating effect. Under 7.9 μm illumination, the light responsivity of the LiNbO<sub>3</sub>-based photodetector with the SiN layer was approximately 600 times higher than that of a conventional graphene photodetector using a Si/SiO<sub>2</sub> substrate. This photodetector takes advantage of the synergistic effect of the pyroelectric and photogating effects, resulting in exceptional performance. This approach provides researchers with a new idea that, in addition to structural design, the synergistic effects of different phenomena during the optoelectronic conversion process can be utilized to induce a photogating effect within the photodetector, thereby improving its performance.

#### 4.3.2. Other 2D materials

In addition to graphene, other 2D materials also have extensive applications in the field of optoelectronic detectors, especially TMDs. As typical 2D materials, TMDs exhibit a large bandgap and moderate carrier mobility, making them ideal for research in electronic and optoelectronic devices [112–115]. Apart from utilizing bare TMDs to fabricate optoelectronic detectors for inducing photogating effects, constructing heterostructures to further modulate the photogating effect is also a commonly employed strategy by researchers.

**4.3.2.1. Bare TMDs.** The photogating effect in optoelectronic detectors based on bare TMDs is typically induced by trap states resulting from defects and impurities inherent in the TMDs themselves. As a result,



**Fig. 6.** (a) Schematic structure of the TiO<sub>2</sub>/MoS<sub>2</sub> heterostructure photodetector [122]; (b) Responsivity of the MoS<sub>2</sub> photodetector and TiO<sub>2</sub>/MoS<sub>2</sub> heterostructure photodetector under different gate source voltages and incident light power densities [122]; (c)  $D^*$  of the MoS<sub>2</sub> photodetector and TiO<sub>2</sub>/MoS<sub>2</sub> heterostructure photodetector under different gate-source voltages and incident light power densities [122]; (d) Schematic structure of the Pbl<sub>2</sub>/WS<sub>2</sub> heterostructure photodetector [123]; (e) Schematic structure of the BP/WSe<sub>2</sub> heterostructure photodetector [125]; (f) Gain and  $D^*$  of the BP/WSe<sub>2</sub> heterostructure photodetector at different wavelengths [125].

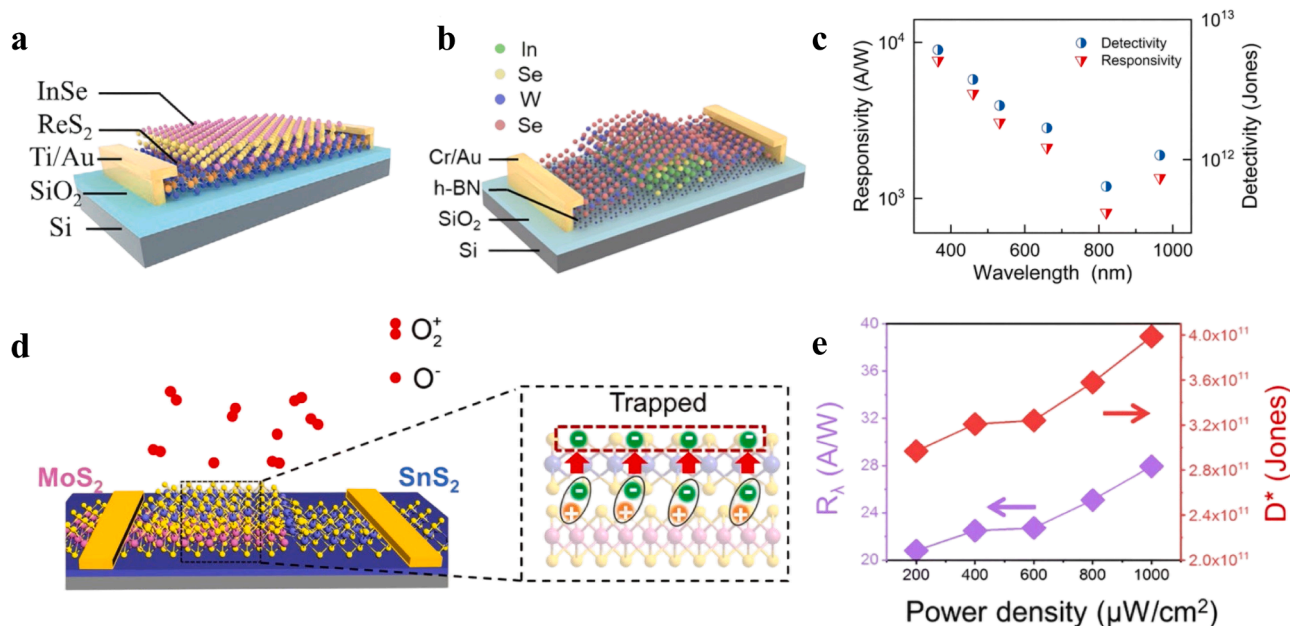
thanks to the photogating effect, optoelectronic detectors based on bare TMDs generally exhibit excellent detection performance. WS<sub>2</sub>, due to its high carrier mobility, favorable bandgap, strong light-matter interaction, and defect-free surface, has been widely applied in optoelectronic detectors. Liu et al. directly grew monolayer WS<sub>2</sub> on a Si<sub>3</sub>N<sub>4</sub> substrate using chemical vapor deposition, forming the structure shown in Fig. 5a [116]. Using scanning transmission electron microscopy and electron energy loss spectroscopy, the researchers discovered various impurity/defect types within different optical subdomains of WS<sub>2</sub>. These defects and impurities act as carrier traps, affecting the electrical and optical emission properties of WS<sub>2</sub>. Meanwhile, these carrier traps also induce a photogating effect, enabling the detector to achieve a responsivity  $>10^5$  under low excitation power (Fig. 5b). Similarly, MoTe<sub>2</sub> has also been widely applied in optoelectronic detectors. Huang et al. fabricated a high-performance optoelectronic detector based on MoTe<sub>2</sub> nanosheets, which achieved an ultra-broad spectral detection range from visible to infrared light (0.6–1.55 μm) and exhibited extremely high optical response characteristics [117]. Through analysis, they found that the photogating effect played a significant role in this high-performance optoelectronic detector. Specifically, when photons are absorbed by MoTe<sub>2</sub>, photogenerated electron-hole pairs are generated, and the tail states of the valence and conduction bands act as trap states to capture holes. These captured holes function as a local gate voltage, increasing the electron density in the channel and modulating its transport, thereby enhancing the photocurrent. The photogating effect improves the device's optical responsivity and detectivity, enabling high photocurrents even under low optical power.

In addition, a unique phenomenon in WSe<sub>2</sub>-based optoelectronic detectors enables simpler utilization of the photogating effect to achieve excellent performance. This phenomenon is the formation of a self-limiting oxide layer, WO<sub>x</sub>, on the surface of WSe<sub>2</sub> under certain conditions. Yamamoto et al. investigated WSe<sub>2</sub>-based optoelectronic detectors, where a self-limiting oxide layer, WO<sub>x</sub>, forms on the surface of WSe<sub>2</sub>, as shown in the schematic diagram in Fig. 5c [118]. WO<sub>x</sub> acts as an effective electron trap in this optoelectronic detector, leading to the

photogating effect. As a result, the photocurrent changes on a timescale of seconds under illumination, with part of the photocurrent persisting after the light is turned off, indicating that photoexcited electrons are deeply trapped by the surface oxide. This effectively modulates the conductivity of the underlying WSe<sub>2</sub>. However, the deeply trapped carriers have difficulty recombining after the light is switched off, resulting in slower response times. Due to the presence of the photogating effect, the responsivity of the WSe<sub>2</sub> transistor with a self-limiting oxide layer, WO<sub>x</sub>, was observed to exceed 3000 A·W<sup>-1</sup> at an incident light power of 1.1 nW. Additionally, Se-based TMDs exhibit unique behaviors in optoelectronic detectors due to the presence of Se vacancies. Yang et al. designed a PtSe<sub>2</sub>-based optoelectronic detector using Se vacancies, as shown in Fig. 5d, achieving an ultra-high responsivity of  $5 \times 10^4$  A·W<sup>-1</sup> (Fig. 5e) [119]. This optoelectronic detector successfully induced the photogating effect through the trap states provided by Se vacancies, and a transition from PPC to NPC was observed under illumination, which effectively corroborated the existence of the photogating effect.

Furthermore, in the research on MoS<sub>2</sub>-based optoelectronic detectors, a novel concept known as the dual photogating effect has been proposed. Zhang et al. studied MoS<sub>2</sub>-based optoelectronic detectors, with the structural diagram shown in Fig. 5f [120]. For clarity, the response under illumination is divided into two stages: the first stage involves the accumulation of photogenerated electrons at the interface of MoS<sub>2</sub>, generating an electric field perpendicular to the interface, similar to the photogating effect in semiconductors; the second stage involves the separation and movement of photogenerated carriers in the substrate under the electric field generated in the first stage. Positive charge accumulates at the MoS<sub>2</sub>/substrate interface, inducing another photogating effect that creates opposite charges in the MoS<sub>2</sub> layer, thus increasing the electron density in MoS<sub>2</sub>. This is the dual photogating effect. In fact, the division of the dual photogating effect into two stages is merely for convenience of description; these two stages occur almost simultaneously, interacting and reinforcing each other, significantly enhancing the performance of the optoelectronic detector. Similarly,





**Fig. 7.** (a) Schematic structure of the InSe/ReS<sub>2</sub> heterojunction photodetector [126]; (b) Schematic structure of the p-WSe<sub>2</sub>/n-InSe heterojunction photodetector [128]; (c) Responsivity and  $D^*$  of the p-WSe<sub>2</sub>/n-InSe heterojunction photodetector at different wavelengths [128]; (d) Schematic structure of the SnS<sub>2</sub>/MoS<sub>2</sub> heterojunction photodetector with O<sub>2</sub> plasma treatment on the SnS<sub>2</sub> surface [129]; (e) Responsivity and  $D^*$  of the SnS<sub>2</sub>/MoS<sub>2</sub> heterojunction photodetector with O<sub>2</sub> plasma treatment on the SnS<sub>2</sub> surface under different incident light power densities [129].

Deng et al. designed a MoS<sub>2</sub>/HfO<sub>2</sub>/Si structure optoelectronic detector, which also successfully realized the dual photogating effect, as shown in Fig. 5 g [121]. When the device is illuminated, both the MoS<sub>2</sub> layer and the Si substrate absorb photons and convert them into photogenerated carriers. These carriers can accumulate at the MoS<sub>2</sub>/HfO<sub>2</sub>/Si interface, as illustrated in Fig. 5 h, modulating the carrier transport in the conductive channel. Furthermore, by adjusting the voltages on these two gates, a tunable bipolar optical responsivity was achieved, ranging from +7000 A·W<sup>-1</sup> (Si bottom gate dominant) to 0 A·W<sup>-1</sup> (balanced) and then to -8000 A·W<sup>-1</sup> (MoS<sub>2</sub> gate dominant), as shown in Fig. 5 i. This feature significantly broadens its potential application scenarios. The presence of the dual photogating effect makes the improvement in the optoelectronic detector's performance more pronounced, and the tunable bipolar optical responsivity further extends its applicability.

**4.3.2.2. Heterostructures.** The photodetectors based on 2D heterostructure bilayers can overcome the bandgap limitation inherent in monolayer 2D materials. However, the extremely low light absorption within the bilayer results in a weak light response. As a result, utilizing the photogating effect to enhance the responsivity of heterojunction photodetectors has become a common strategy among researchers. In heterostructures, trap states often arise from charge exchange or charge accumulation at the heterojunction, and these trap states are generally more controllable than those induced by defects or impurities within the 2D materials. Therefore, the photogating effect within heterostructure-based photodetectors exhibits greater tunability, making it easier to balance the relationship between responsivity and response time, thus expanding the application range of photodetectors.

In the selection of materials for constructing heterostructures, some researchers have chosen materials distinct from TMDs to compensate for the inherent drawbacks of TMDs, aiming to significantly improve the performance of photodetectors. Additionally, the establishment of heterostructures makes it easier to manipulate the photogating effect to regulate photodetector performance. Liu et al. deposited an ultrathin TiO<sub>2</sub> layer on several layers of MoS<sub>2</sub> using electron beam evaporation, constructing a TiO<sub>2</sub>/MoS<sub>2</sub> heterojunction photodetector (Fig. 6 a) [122]. Benefiting from the charge transfer at the heterojunction interface and

the photogating effect, the responsivity and  $D^*$  of this photodetector were significantly enhanced compared to pure MoS<sub>2</sub> photodetectors (Fig. 6 b and Fig. 6 c): under visible light illumination and a 60 V gate voltage, the responsivity reached 2199 A·W<sup>-1</sup>, 4.0 times higher than that of the pure MoS<sub>2</sub> photodetector; at an incident light power density of 23.2 μW/mm<sup>2</sup>, the  $D^*$  at zero gate voltage reached  $1.67 \times 10^{13}$  Jones, 4.2 times greater than that of the pure MoS<sub>2</sub> photodetector. Similarly, Qi et al. constructed a p-PbI<sub>2</sub>/n-WS<sub>2</sub> heterojunction photodetector (Fig. 6 d) based on chemical vapor deposition [123]. The strong built-in electric field at the p-n heterojunction resulted in the generation of the photogating effect, significantly suppressing the recombination of photo-generated electron-hole pairs. As a result, the device achieved a  $D^*$  of  $4.9 \times 10^{13}$  Jones and a responsivity of  $5.57 \times 10^2$  A·W<sup>-1</sup>. Moreover, with the adjustment of the gate voltage, the responsivity of the device could be further enhanced: at a -60 V negative gate voltage, the responsivity reached  $7.1 \times 10^4$  A·W<sup>-1</sup>. In addition, through the regulation of the heterojunction, the device also achieved fast rise and decay times of 24 ms and 33 ms, respectively, significantly broadening the application scenarios of photodetectors. BP, a layered crystal structure where phosphorus atoms are covalently bonded to three adjacent atoms within each layer, forms an orthogonal lattice. Its bandgap is dependent on the number of layers, a characteristic that allows BP photodetectors to extend their detection range from visible light to the mid-infrared spectrum, making them potentially useful for infrared photodetector applications [124]. Ye et al. constructed a BP/WSe<sub>2</sub> vertical heterostructure photodetector (Fig. 6 e), where BP serves as the photogating layer and WSe<sub>2</sub> as the conductive channel [125]. Due to the unique properties of BP, the photodetector's detection range covers both visible and infrared light, with responsivities reaching  $10^3$  A·W<sup>-1</sup> for visible light and  $0.5$  A·W<sup>-1</sup> for infrared light. The device also maintains high gain and  $D^*$  over a wide wavelength range (Fig. 6 f).

In addition to using different 2D materials to construct heterostructures with TMDs, some researchers have also used two types of TMDs to fabricate photodetectors, achieving significant performance improvements. Ma et al. designed an InSe/ReS<sub>2</sub> heterostructure photodetector (Fig. 7 a), which realized high sensitivity and a wide spectral response range (365–965 nm) [126]. The top layer, InSe, serves as the



**Table 2**

Figures-of-merit in typical low dimensional photodetectors.

| System                    | Structure                                       | Gain               | Responsivity ( $A \cdot W^{-1}$ )          | Specific detectivity (Jones) | Ref.  |
|---------------------------|-------------------------------------------------|--------------------|--------------------------------------------|------------------------------|-------|
| Quantum Dots              | PbS QDs/InGaZnO                                 | —                  | $>10^6$                                    | $10^{13}$                    | [81]  |
|                           | PbS QDs/graphene                                | $10^8$             | $10^7$                                     | $7 \times 10^{13}$           | [82]  |
|                           | PbS QDs/ graphene                               | —                  | $10^8$                                     | —                            | [83]  |
|                           | Si QDs/MoS <sub>2</sub>                         | —                  | $6.0 \times 10^3$                          | $10^{14}$                    | [84]  |
|                           | PbS QDs/TiO <sub>2</sub> film/graphene          | —                  | $1.2 \times 10^4$                          | $1.5 \times 10^{12}$         | [85]  |
| Nanowires                 | InAs                                            | $10^3$             | $>10^3$                                    | —                            | [88]  |
|                           | InAs                                            | —                  | $4.4 \times 10^3$                          | $2.6 \times 10^{11}$         | [89]  |
|                           | BiSeI                                           | —                  | $7 \times 10^4$                            | $2.5 \times 10^{14}$         | [90]  |
|                           | InAs                                            | $10^5$             | —                                          | —                            | [91]  |
|                           | ZnO NWs/graphene                                | —                  | $1.87 \times 10^5$                         | —                            | [93]  |
| Two-dimensional materials | Graphene/graphene                               | —                  | $10^3$                                     | —                            | [97]  |
|                           | Graphene/Si                                     | —                  | $>10^6$                                    | —                            | [105] |
|                           | Graphene/ReS <sub>2</sub>                       | —                  | $7 \times 10^5$                            | —                            | [106] |
|                           | Graphene/BP                                     | $1.13 \times 10^9$ | $3.3 \times 10^3$                          | —                            | [107] |
|                           | MoTe <sub>2</sub> /graphene                     | $4.69 \times 10^8$ | 970.82                                     | $1.55 \times 10^{11}$        | [108] |
|                           | Graphene/SiO <sub>2</sub> /Si                   | —                  | $>10^3$                                    | —                            | [109] |
|                           | Graphene/Si                                     | —                  | 240                                        | $3.4 \times 10^{12}$         | [110] |
|                           | WS <sub>2</sub> /Si <sub>3</sub> N <sub>4</sub> | —                  | $>10^5$                                    | —                            | [116] |
|                           | WO <sub>x</sub> /WSe <sub>2</sub>               | —                  | $3 \times 10^3$                            | —                            | [118] |
|                           | PtSe <sub>2</sub>                               | —                  | $5 \times 10^4$                            | —                            | [119] |
|                           | MoS <sub>2</sub> /HfO <sub>2</sub> /Si          | —                  | $-8000 \sim +7000$                         | —                            | [121] |
|                           | TiO <sub>2</sub> /MoS <sub>2</sub>              | —                  | 2199                                       | $1.67 \times 10^{13}$        | [122] |
|                           | PbI <sub>2</sub> /WS <sub>2</sub>               | —                  | $7.1 \times 10^4$                          | $4.9 \times 10^{13}$         | [123] |
|                           | BP/WSe <sub>2</sub>                             | $10^3$             | $>10^5$                                    | $>10^{14}$                   | [125] |
|                           | InSe/ReS <sub>2</sub>                           | $10^6$             | 1921                                       | $6.51 \times 10^{13}$        | [126] |
|                           | MoS <sub>2</sub> /MoSe <sub>2</sub>             | —                  | 680                                        | —                            | [127] |
|                           | WSe <sub>2</sub> /InSe                          | —                  | $-1.76 \times 10^4 \sim -5.48 \times 10^4$ | —                            | [128] |
|                           | SnS <sub>2</sub> /MoS <sub>2</sub>              | —                  | 24                                         | $>4 \times 10^{11}$          | [129] |

photogating layer to modulate the photogenerated current in the channel, achieving a gain of up to  $10^6$ , a responsivity of  $1921 A \cdot W^{-1}$ , and a  $D^*$  of  $6.51 \times 10^{13}$  Jones. Li et al. constructed a MoS<sub>2</sub>/MoSe<sub>2</sub> heterostructure photodetector, which exhibited a high responsivity of  $680 A \cdot W^{-1}$  under the regulation of the photogating effect [127]. Lei et al. observed tunable bipolar photoresponsivity (from  $-1.76 \times 10^4$  to  $5.48 \times 10^4 A \cdot W^{-1}$ ) in a p-WSe<sub>2</sub>/n-InSe heterostructure photodetector (Fig. 7b) [128]. Moreover, this photodetector maintained ultra-high responsivity and  $D^*$  across a wide spectral range of 400–1000 nm (Fig. 7c). Beyond the simple construction of heterostructures with two types of TMDs, and utilizing the trapping effect at the heterojunction to induce the photogating effect, new strategies have also been proposed. Quan et al. constructed an SnS<sub>2</sub>/MoS<sub>2</sub> heterojunction photodetector and treated the surface of SnS<sub>2</sub> with O<sub>2</sub> plasma, introducing a large number of trap states [129]. This enabled a tunable photogating effect (Fig. 7d): under light illumination, photogenerated electrons were trapped in the trap states on the SnS<sub>2</sub> surface, modulating the conduction of photogenerated holes in the 2D material layers. By controlling the O<sub>2</sub> plasma treatment method, they could indirectly control the strength of the photogating effect in the device, balancing the responsivity and response time. Thanks to the photogating effect, this photodetector achieved extremely high responsivity and  $D^*$  under various incident light power densities (Fig. 7e). Due to the photogating effect, the photodetector achieves a responsivity of  $26 A \cdot W^{-1}$  and a  $D^*$  of  $4.0 \times 10^{11}$  Jones under an incident light power density of  $1000 \mu W/cm^2$  (Fig. 7f).

In conclusion, low-dimensional materials, such as quantum dots, nanowires, and two-dimensional materials, provide an ideal platform for the realization of the photogating effect due to their unique optoelectronic properties and structural characteristics. By incorporating designs such as defect states and heterojunctions, these materials can effectively induce the photogating effect, significantly enhancing the gain, responsivity, and  $D^*$  of photodetectors (Table 2). However, despite the performance improvements brought about by the photogating effect, challenges remain in balancing response speed and sensitivity. With continuous optimization of material and structural designs, low-dimensional photodetectors hold great potential for high-performance detectors in the future, particularly in fields such as high sensitivity,

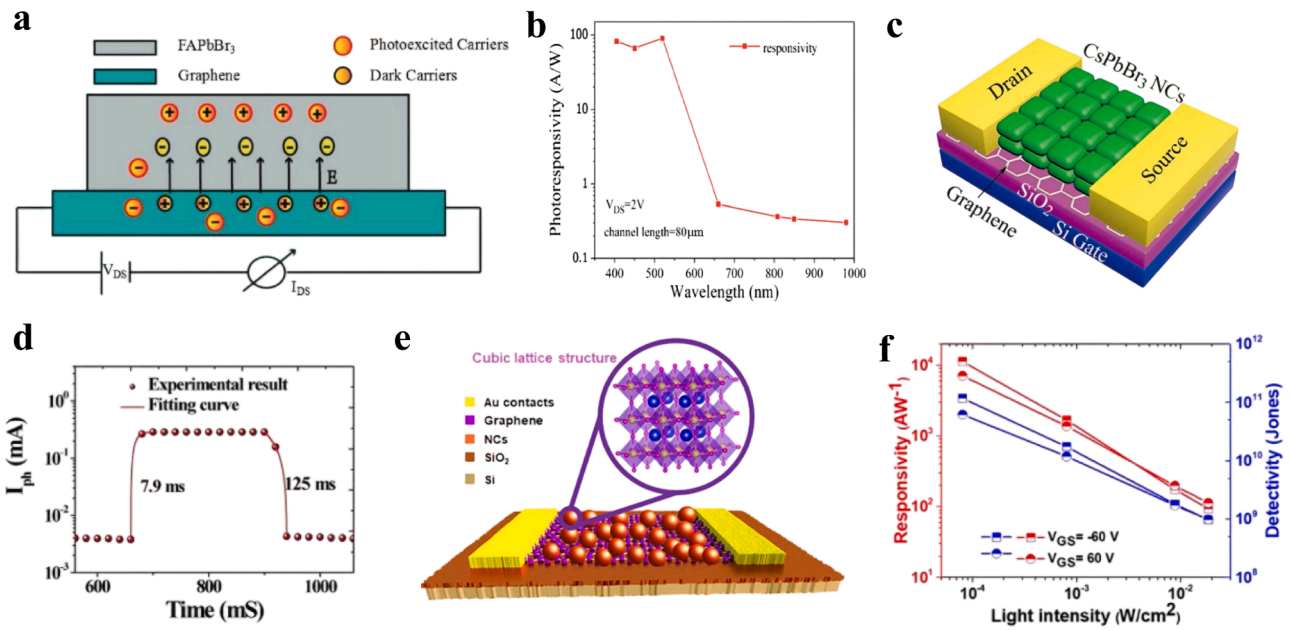
broad spectral response, and high-speed imaging.

## 5. Photogating effect in perovskite-based hybrid photodetectors

As a direct-bandgap semiconductor with a high absorption coefficient and high quantum efficiency, perovskite has demonstrated exceptional performance not only in solar cells but also in the field of photodetectors in recent years [130–140]. Owing to its unique optoelectronic properties, particularly its efficient light absorption across the ultraviolet to near-infrared spectrum and excellent carrier mobility, perovskite holds great potential for photodetector applications. Researchers have leveraged the exceptionally high absorption coefficient of perovskite thin films, incorporating them as the light-absorbing layer in heterostructures with other materials [141]. Notably, integrating perovskite with conventional 2D materials can significantly compensate for the inherently poor light absorption capability of 2D materials [142]. Moreover, due to the inherently high defect density of perovskites, certain defects in their low-dimensional forms can introduce trap states within the band structure, thereby inducing the photogating effect [143–145]. Consequently, in heterostructure design, some researchers have explored the integration of low-dimensional perovskite with other materials. The combination of its high absorption coefficient and unique low-dimensional material properties allows low-dimensional perovskite to meet different requirements for  $D^*$  in photodetectors [146].

### 5.1. Perovskite/graphene heterostructures

Perovskite is often combined with typical 2D materials, such as graphene, to construct heterojunction photodetectors, thereby inducing the photogating effect. Perovskite/graphene heterostructure photodetectors combine the excellent properties of both materials, demonstrating tremendous potential in the field of photodetection. Graphene, as a 2D material, possesses ultra-high carrier mobility, a broad spectral response range, and ultrafast photoresponse speed. However, its intrinsic light absorption rate is only 2.3 %, limiting its effective utilization of incident light and, consequently, its photodetection performance [147–149]. In contrast, perovskite materials exhibit an extremely

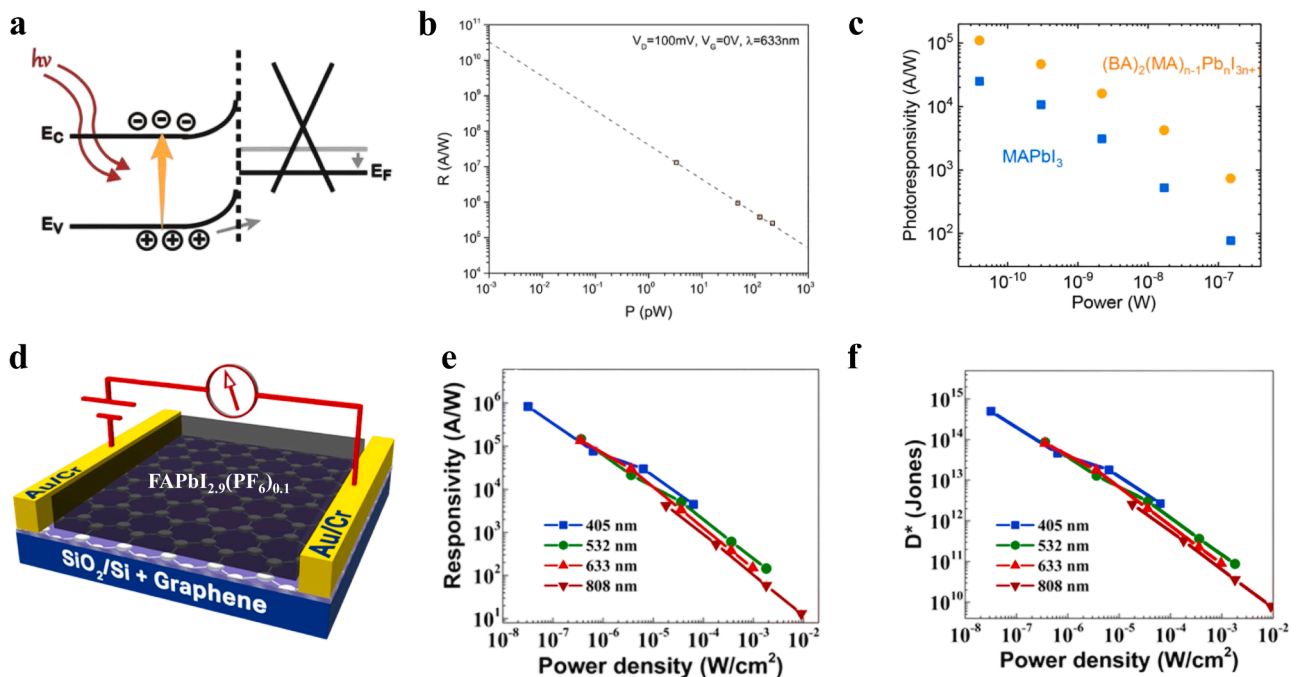


**Fig. 8.** (a) Schematic diagram of the photogating effect principle in FAPbBr<sub>3</sub> QDs/graphene heterojunction photodetector [155]; (b) Responsivity of FAPbBr<sub>3</sub> QDs/graphene photodetector at different wavelengths [155]; (c) Schematic structure of CsPbBr<sub>3</sub> nanocrystal/graphene heterojunction photodetector [156]; (d) Temporal  $I_{ph}$  response during one on-off illumination switching [156]; (e) Schematic structure of CsPbBr<sub>3</sub>I<sub>3-x</sub>/graphene heterojunction photodetector [157]; (f) Responsivity and  $D^*$  of CsPbBr<sub>3</sub>I<sub>3-x</sub>/graphene heterojunction photodetector under two gate source voltages and varying incident light power densities [157].

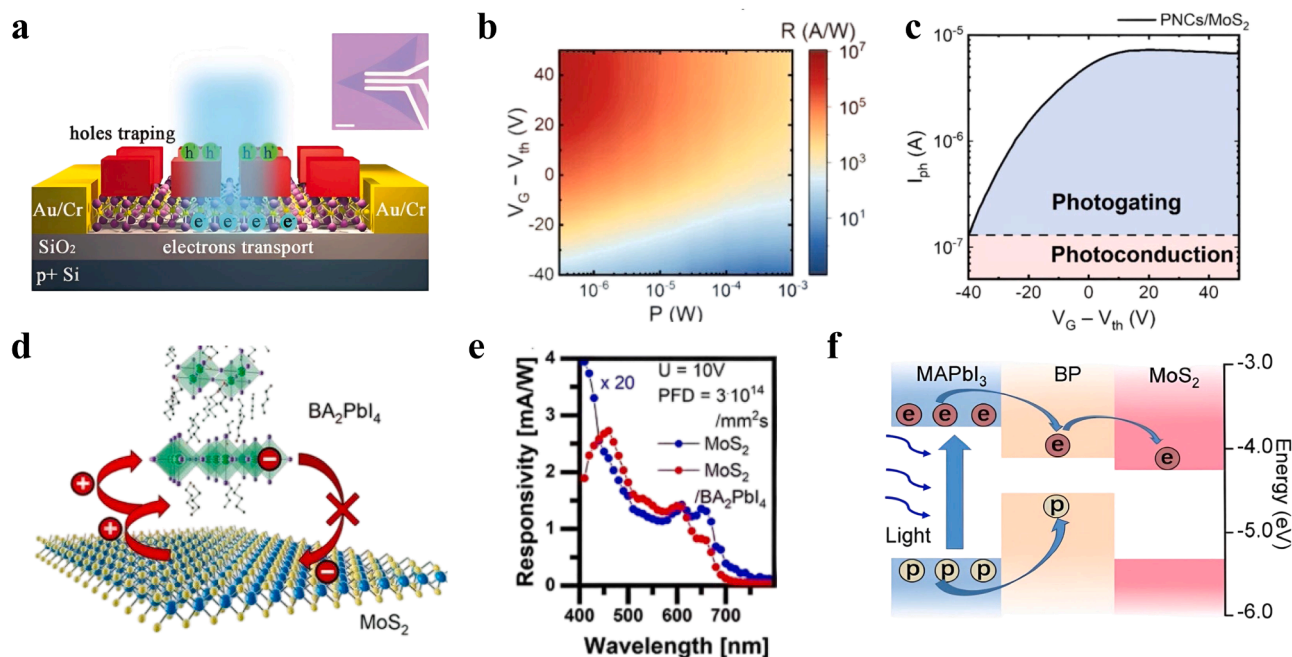
high absorption coefficient, enabling even low-dimensional perovskite to efficiently absorb visible and near-infrared light, generating high concentrations of photogenerated carriers [150,151]. Therefore, in perovskite/graphene heterostructures, perovskite can serve as an efficient light-absorbing layer, compensating for the weak absorption capability of graphene and enhancing the overall optoelectronic conversion efficiency of the photodetector. Furthermore, the good energy

level alignment between graphene and perovskite allows for efficient transfer of photogenerated electrons to the graphene layer. The high carrier mobility of graphene aids in accelerating carrier transport, thereby improving the photodetector's response speed and sensitivity [149,152].

Compared to 1D and 2D perovskites, perovskite QDs retain more high light absorption while exhibiting the characteristics of low-



**Fig. 9.** (a) Band diagram of the nanowires/graphene heterojunction for gate voltages ( $V_G$ ) lower than the charge neutrality point [158]; (b) Responsivity of perovskite nanowire/graphene heterojunction photodetector under different light excitation power [158]; (c) The contrast of responsivity before and after the introduction of BA molecules into MAPbI<sub>3</sub> [159]; (d) Schematic structure of perovskite/graphene heterojunction photodetector [161]; (e) Responsivity of perovskite/graphene heterojunction photodetector under different wavelengths and incident light power densities [161]; (f)  $D^*$  of perovskite/graphene heterojunction photodetector under different wavelengths and incident light power densities [161].

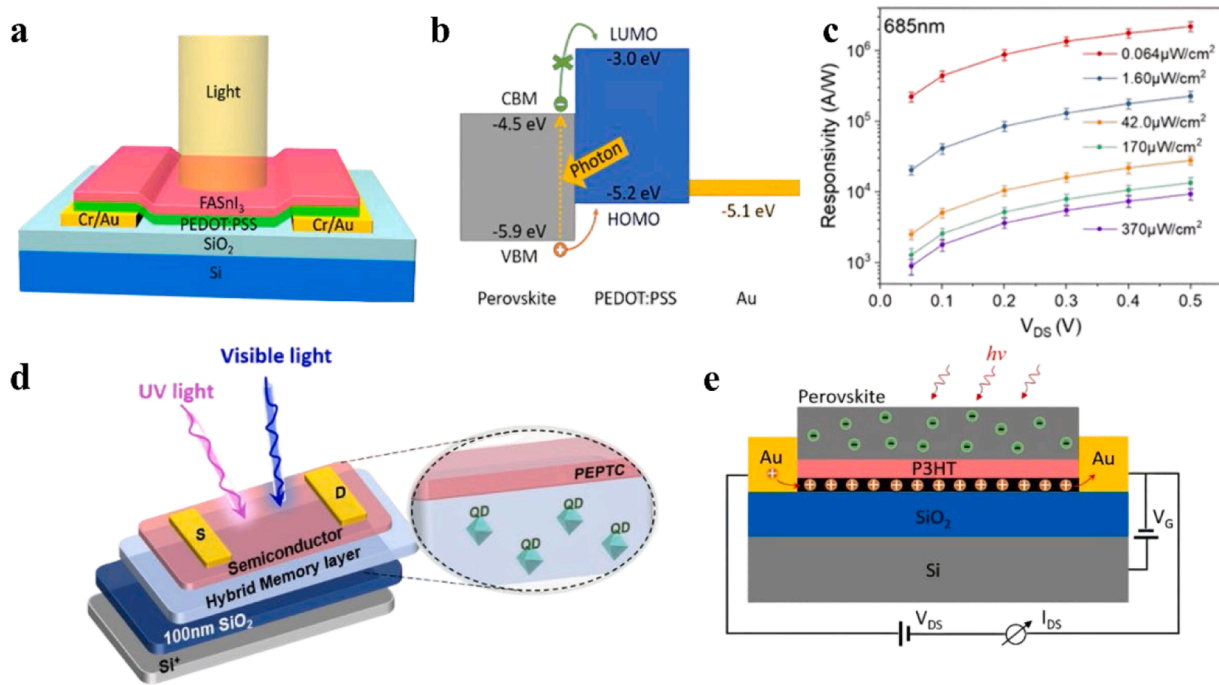


**Fig. 10.** (a) Schematic of the perovskite quantum dots/MoS<sub>2</sub> heterojunction photodetector structure [162]; (b) Schematic of the responsivity distribution of the in-situ core-shell structure perovskite/MoS<sub>2</sub> heterojunction photodetector under different gate voltages and incident light power [163]; (c) Gate dependence of the photocurrent [163]; (d) Schematic of the island-like BA<sub>2</sub>PbI<sub>4</sub> perovskite/MoS<sub>2</sub> heterojunction photodetector structure [166]; (e) Responsivity of the MoS<sub>2</sub> photodetector and the island-like BA<sub>2</sub>PbI<sub>4</sub> perovskite/MoS<sub>2</sub> heterojunction photodetector under different wavelengths [166]; (f) Schematic of the MAPbI<sub>3</sub>/BP/MoS<sub>2</sub> energy levels and the migration of carriers under light illumination conditions [167].

dimensional materials. Zheng et al. deposited CsPbBr<sub>3</sub> QDs on a graphene-based photodetector [153]. Under light irradiation, photo-generated electron-hole pairs are generated and separated in the perovskite QDs. Holes are transported from the QDs into the graphene layer, while photogenerated electrons are captured by the perovskite QDs, forming a local gate voltage that regulates hole transport, thus inducing a photogating effect. Thanks to the strong light absorption of the perovskite material, the high mobility of graphene, and the regulation of the photogating effect, the device achieves a responsivity of  $22.8 \text{ A} \cdot \text{W}^{-1}$  at a wavelength of 266 nm, with a  $D^*$  of  $1.89 \times 10^{10}$  Jones. Similarly, Li et al. sequentially deposited graphene and perovskite QDs on paper-based substrates, and the photogenerated electrons were similarly trapped in the perovskite QDs to generate the photogating effect [154]. At a wavelength of 520 nm, the responsivity reached  $8.2 \times 10^4 \text{ A} \cdot \text{W}^{-1}$ . Benefiting from the broad spectral absorption of graphene materials, Pan et al. fabricated a perovskite QDs/graphene heterojunction photodetector, which achieved a broad spectral response range from 405 to 980 nm, as shown in Fig. 8a [155]. In this device, the photogating effect operates under light irradiation. The device exhibits higher responsivity in the 405–538 nm range, while the 538–980 nm range shows lower responsivity but faster response times (Fig. 8b). The device achieves an ultrahigh responsivity of  $1.15 \times 10^5 \text{ A} \cdot \text{W}^{-1}$  and an EQE of  $3.42 \times 10^7 \%$  under an incident light power of  $3 \mu\text{W}$  at a wavelength of 520 nm. In addition to perovskite QDs, slightly larger perovskite nanocrystals are also commonly used in photodetectors to construct heterostructures with graphene. Che et al. deposited CsPbBr<sub>3</sub> nanocrystals on graphene to fabricate a perovskite nanocrystal/graphene heterojunction photodetector (Fig. 8c) [156]. The nanocrystals' trapping effect on photogenerated electrons led to the photogating effect, significantly enhancing the device's light response performance: under 405 nm light illumination, the responsivity,  $D^*$ , and EQE reached  $3.4 \text{ A} \cdot \text{W}^{-1}$ ,  $7.5 \times 10^8$  Jones, and  $10^3 \%$ , respectively. In addition, due to the high mobility within graphene, the device has achieved a rather fast response time, reaching the millisecond level (Fig. 8d). Similarly, Surendran et al. constructed a CsPbBr<sub>3</sub>I<sub>3-x</sub>

nanocrystal/graphene heterostructure (Fig. 8e) [157]. Under the regulation of the photogating effect, this device maintained extremely high responsivity and  $D^*$  at both  $-60 \text{ V}$  and  $+60 \text{ V}$  gate voltages (Fig. 8f).

In addition to perovskite QDs and similar nanocrystals, other types of perovskites can also be used to construct heterojunction photodetectors with graphene. Spina et al. combined perovskite nanowires with monolayer graphene to build a heterojunction photodetector [158]. The schematic diagram of the energy band structure of this structure is shown in Fig. 9a. The built-in field formed at the heterojunction promotes the injection of photogenerated holes into graphene. The electrons accumulated in the perovskite nanowires generate an electric field, which further regulates the carriers in graphene. Under light irradiation, the carrier trapping effect of perovskite induced the photogating effect, enabling the device to maintain extremely high responsivity over a wide wavelength range (Fig. 9b). In addition to nanowires, 2D perovskites are also commonly used in perovskite/graphene heterostructures. Shao et al. constructed a 2D perovskite/graphene heterojunction photodetector [159]. In this article, by introducing BA molecules into MAPbI<sub>3</sub> molecules, the regulation of the performance of perovskite has been achieved. Compared with MAPbI<sub>3</sub>, the performance of the device has been significantly improved (Fig. 9c). Due to the band bending at the heterojunction, electrons transfer from perovskite to graphene at the interface, while holes accumulate in the perovskite, inducing the photogating effect, ultimately resulting in a high responsivity of  $10^5 \text{ A} \cdot \text{W}^{-1}$  at a wavelength of 532 nm. Apart from low-dimensional perovskites, perovskite thin films can also be used to construct heterojunction photodetectors with graphene. Compared to low-dimensional perovskites, although perovskite thin films lose some of the characteristics of low-dimensional materials, their sufficient thickness ensures an extremely high light absorption coefficient. Liu et al. fabricated a perovskite/graphene heterojunction photodetector, where perovskite served as the light-absorbing layer, compensating for the weak light absorption of graphene [160]. The photogating effect at the heterojunction enabled the device to achieve a responsivity of over  $10^4 \text{ A} \cdot \text{W}^{-1}$ . It is evident that the properties of perovskite themselves have a



**Fig. 11.** (a) Schematic diagram of the perovskite/PEDOT:PSS heterojunction photodetector structure [172]; (b) Schematic diagram of the energy band alignment and carrier transport under light irradiation for the perovskite/PEDOT:PSS heterojunction photodetector [172]; (c) Responsivity of the perovskite/PEDOT:PSS heterojunction photodetector at 685 nm under different incident light power densities [172]; (d) Perovskite QDs/polymer heterojunction photodetector [176]; (e) Schematic diagram of the perovskite/P3HT/graphene photodetector structure [177].

significant impact on the performance of perovskite/graphene photodetectors. Therefore, improving the quality of perovskite thin films is one of the key strategies for researchers to enhance the performance of perovskite-based photodetectors. Sridhar et al. partially replaced the I in FAPbI<sub>3</sub> with the superhalide PF<sub>6</sub>, improving the stability of perovskite, enhancing its light absorption coefficient, and used this perovskite thin film to construct a heterojunction photodetector with graphene (Fig. 9d) [161]. Under the regulation of the photogating effect, the device achieved a responsivity exceeding  $10^6 \text{ A} \cdot \text{W}^{-1}$  and a  $D^*$  exceeding  $10^{14}$  Jones at a wavelength of 405 nm (Fig. 9e and Fig. 9f).

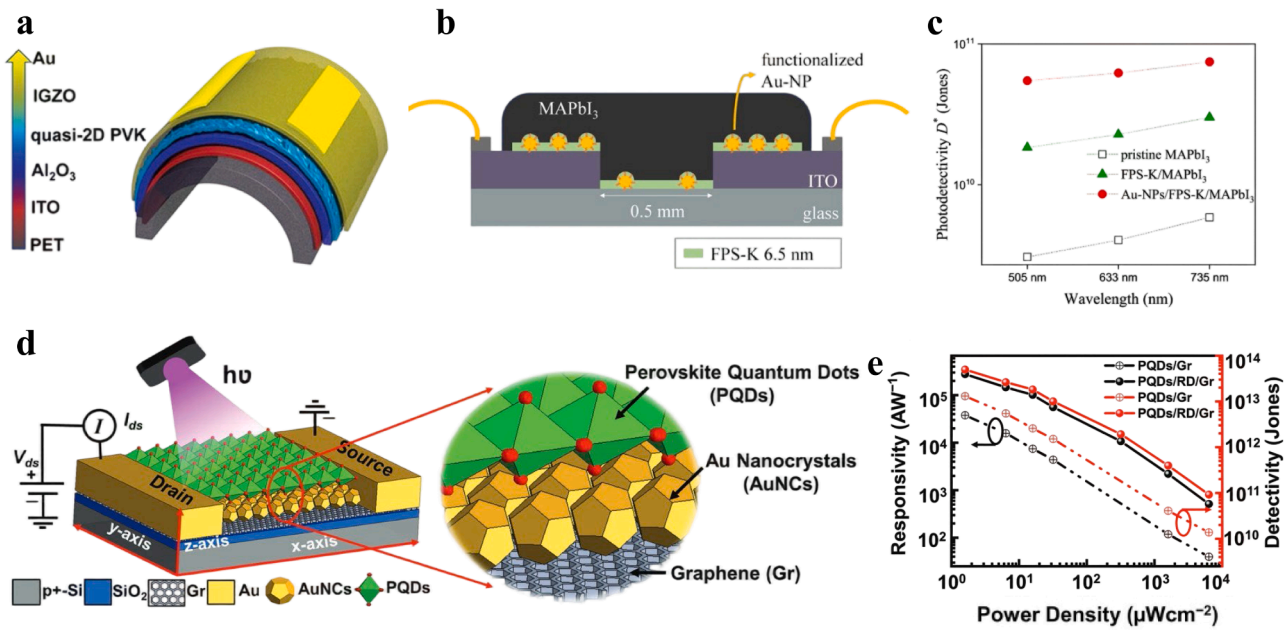
## 5.2. Perovskite/TMDs heterostructures

TMDs, as one of the representative two-dimensional materials, have also been widely combined with perovskites to construct high-performance photodetectors. MoS<sub>2</sub>, as a typical TMDs, has been combined with perovskite QDs by Wu et al. [162]. Compared to 1D and 2D perovskites, perovskite QDs retain more high light absorption while maintaining low-dimensional material characteristics. Under light illumination, photogenerated holes are trapped in the perovskite QDs, while electrons are transported in MoS<sub>2</sub>, as shown in Fig. 10a. Due to the presence of the photogating effect, the device exhibits excellent performance: light gain  $>10^5$ , light responsivity of  $5.5 \times 10^4 \text{ A} \cdot \text{W}^{-1}$ , and  $D^*$  of  $1.95 \times 10^{12}$  Jones. Similarly, Sim et al. combined 0D perovskite (in situ core-shell structure perovskite) with 2D MoS<sub>2</sub> to construct a photodetector [163]. In this structure, perovskite captures photogenerated holes, while photogenerated electrons are transferred into MoS<sub>2</sub>. The extended hole lifetime in the perovskite leads to a significant increase in the photogenerated current of the device. The device maintains a high responsivity under different gate voltages and incident light power (Fig. 10b). It is worth mentioning that, compared with the photoconductive effect, the photogating effect has a stronger dependence on the gate electrode, and it is easier to be regulated under the gate voltage to meet different application scenarios (Fig. 10c). In both these perovskite QD/2D material heterojunction photodetectors, the performance of the

QDs themselves significantly influences the photogating effect, thereby affecting the overall performance of the photodetector. Therefore, some researchers have focused on how to regulate perovskite QDs to optimize the performance of photodetectors. Han et al. studied a unique discrete phase-transition method to design tunable CsPbBr<sub>3</sub>/Cs<sub>4</sub>PbBr<sub>6</sub> quantum dots in Cs<sub>4</sub>PbBr<sub>6</sub> nanocrystals and applied them to photodetectors [164]. Compared to single CsPbBr<sub>3</sub> quantum dots, photodetectors based on CsPbBr<sub>3</sub>/Cs<sub>4</sub>PbBr<sub>6</sub> quantum dots exhibit higher photocurrents under different gate voltages, indicating that the hybrid quantum dots induce the photogating effect more effectively. However, regardless of how the QDs are engineered, their inherently low carrier mobility remains unavoidable, which is attributed to the disorder induced by traditional quantum dot fabrication methods. To address this issue, Luo et al. aligned CsPbBr<sub>3</sub> QDs to form superlattices [165]. The coupling effects within the superlattice impart quasi-2D characteristics, resulting in a significant enhancement of carrier mobility. When this superlattice was integrated with WS<sub>2</sub> to fabricate a photodetector, it achieved a responsivity of  $91.24 \text{ A} \cdot \text{W}^{-1}$  and a  $D^*$  of  $1.15 \times 10^{11}$  Jones. Notably, the quasi-2D nature of the superlattice enabled exceptionally fast carrier mobility, leading to a rapid response time of 160 μs/380 μs for the photodetector.

In addition to using low dimensional perovskite, some researchers have also used perovskite thin film to construct heterostructures with MoS<sub>2</sub>. Spee et al. introduced an island-like BA<sub>2</sub>PbI<sub>4</sub> perovskite layer on a monolayer MoS<sub>2</sub> [166]. The photogenerated holes from MoS<sub>2</sub> are captured and guided to the BA<sub>2</sub>PbI<sub>4</sub> layer, thereby inducing the photogating effect (Fig. 10d). The study confirmed that the heterostructure allows only holes to transfer from the MoS<sub>2</sub> layer to the BA<sub>2</sub>PbI<sub>4</sub> layer, inhibiting any other possible charge transfer pathways. Compared to pure MoS<sub>2</sub> photodetectors, the introduction of the perovskite layer greatly improved the responsivity of the device (Fig. 10e). This excellent performance is attributed not only to the photogating effect but also to the high light absorption coefficient of the perovskite. To achieve better energy level alignment for efficient carrier transport and photogating effect modulation, Wang et al. inserted BP between the perovskite and





**Fig. 12.** (a) Schematic illustration of the quasi-2D perovskite/InGaZnO heterojunction phototransistor structure [178]; (b) Schematic illustration of the perovskite/AuNPs/FPS-K photodetector structure [179]; (c) Comparison of the  $D^*$  of photodetectors based on pure perovskite, perovskite/FPS-K, and perovskite/AuNPs/FPS-K structures at different wavelengths [179]; (d) Schematic illustration of the perovskite QDs/AuNCs/graphene photodetector structure [180]; (e) Comparison of the responsivity and  $D^*$  of perovskite QDs/graphene and perovskite QDs/RD/graphene structures under different incident light power densities [180].

MoS<sub>2</sub> layers, resulting in better energy level matching. Under light illumination, the perovskite layer serves as the light absorption layer generating photogenerated electron-hole pairs [167]. With good energy level alignment, the photogenerated electrons are transferred from the perovskite layer through the BP layer into the MoS<sub>2</sub> layer, while photogenerated holes accumulate in the BP, making the BP layer act as the photogating layer and modulating the transport of photogenerated electrons in the MoS<sub>2</sub> layer (Fig. 10f). This photodetector combines the advantages of BP, MoS<sub>2</sub>, and the high light absorption of perovskite. Under reverse bias conditions, the device exhibits high responsivity (11 A·W<sup>-1</sup>), high  $D^*$  ( $1.3 \times 10^{12}$  Jones), and fast response times (rise/decay times of 150/240 μs).

### 5.3. Perovskite/organic material heterostructures

The perovskite/organic material heterojunction photodetector combines the advantages of both materials, offering significant improvements, particularly in terms of optoelectronic performance and manufacturability. Compared to inorganic materials such as graphene and TMDs, organic materials offer advantages such as low cost, flexibility, and tunable energy band structures, making them conducive to large-scale production and flexible applications [168,169]. Perovskite, known for their excellent light absorption capabilities, can form a heterostructure when combined with organic materials. This combination improves carrier transport through complementary energy band alignment, enhances optoelectronic response, and utilizes organic materials to increase the environmental stability of the device, optimizing the performance of the photodetector [170,171]. This structure not only improves the detector's sensitivity and response speed but also expands its application areas, such as in flexible electronics and optoelectronic sensors.

Poly-(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is a classic organic semiconductor material. Liu et al. fabricated a perovskite/PEDOT:PSS heterojunction photodetector (Fig. 11a) [172]. Due to energy level differences, electrons are trapped at the heterojunction interface and have difficulty recombining with holes, while holes are transported multiple times in the PEDOT:PSS channel with

high hole mobility (Fig. 11b), significantly increasing the photocurrent. By combining the advantages of organic materials and perovskite along with the photogating effect, the device achieved an exceptionally high responsivity at 685 nm (Fig. 11c). Similarly, Gu et al. and Xie et al. also constructed perovskite/PEDOT:PSS heterojunction photodetectors, and through the photogating effect and the complementary properties of both materials, the devices achieved excellent performance [173,174]. In particular, the photodetector fabricated by Xie et al. exhibited a high responsivity of about  $10^9$  A·W<sup>-1</sup> and a  $D^*$  of  $10^{14}$  Jones across a wide UV to near-infrared bandwidth. In addition to PEDOT:PSS, [6,6]-phenyl-C61-butyric acid methyl ester (PCBM) is also a commonly used organic semiconductor material. Qin et al. demonstrated that introducing 1 % PCBM into perovskite photodetectors could increase their responsivity to  $8 \times 10^5$  A·W<sup>-1</sup>, which is approximately 30 times higher than that of pure perovskite devices [175].

In addition to simple photodetector applications, perovskite/organic material heterojunction photodetectors and the photogating effect have also been widely applied in memory devices. Chiang et al. fabricated a polymer/perovskite QDs heterojunction photodetector (Fig. 11d) [176]. By utilizing the photogating effect between the UV-sensitive polymer and the visible-light-sensitive quantum dots, they achieved a fully optically switchable memory device, resulting in excellent optical responsiveness and memory switchability.

Some researchers have also constructed multi-heterojunction photodetectors to enhance the performance of photodetectors. Xie et al. introduced poly (3-hexylthiophene) (P3HT) into the perovskite/graphene heterojunction photodetector, placing it between the perovskite and graphene layers (Fig. 11e) [177]. Photogenerated electron-hole pairs are effectively separated in P3HT and transferred to the perovskite and graphene layers, with high-density electrons being trapped in the perovskite layer, inducing the photogating effect. This structure achieved an exceptionally high responsivity of  $4.3 \times 10^9$  A·W<sup>-1</sup> and a nearly  $10^{10}$  gain. Furthermore, the device demonstrated detection capabilities across an extensive ultraviolet to near-infrared spectrum, showing great potential for future applications.

**Table 3**  
Figures-of-merit in typical perovskite photodetectors.

| Structures                             | Gain      | Responsivity<br>( $A \cdot W^{-1}$ ) | Specific<br>detectivity<br>(Jones) | Ref.  |
|----------------------------------------|-----------|--------------------------------------|------------------------------------|-------|
| Perovskite QDs/<br>graphene            | —         | 22.8                                 | $1.89 \times 10^{10}$              | [153] |
| Perovskite QDs/<br>graphene            | —         | $8.2 \times 10^4$                    | —                                  | [154] |
| Perovskite QDs/<br>graphene            | —         | $1.15 \times 10^5$                   | —                                  | [155] |
| Perovskite QDs/<br>graphene            | —         | 3.4                                  | $7.5 \times 10^8$                  | [156] |
| Perovskite<br>nanocrystal/<br>graphene | —         | $>10^4$                              | $>10^{11}$                         | [157] |
| Perovskite nanowires/<br>graphene      | —         | $>10^{10}$                           | —                                  | [158] |
| 2D perovskite/<br>graphene             | —         | $10^5$                               | —                                  | [159] |
| Perovskite/graphene                    | —         | $>10^4$                              | —                                  | [160] |
| Perovskite/graphene                    | —         | $10^6$                               | —                                  | [161] |
| Perovskite QDs/MoS <sub>2</sub>        | $>10^5$   | $5.5 \times 10^4$                    | $1.95 \times 10^{12}$              | [162] |
| 0D perovskite/MoS <sub>2</sub>         | —         | $10^7$                               | —                                  | [163] |
| Perovskite QDs/WS <sub>2</sub>         | —         | 91.24                                | $1.15 \times 10^{11}$              | [165] |
| Perovskite/BP/MoS <sub>2</sub>         | —         | 11                                   | $1.3 \times 10^{12}$               | [167] |
| Perovskite/PEDOT:<br>PSS               | —         | $>10^6$                              | —                                  | [172] |
| Perovskite/PEDOT:<br>PSS               | —         | $10^9$                               | $10^{14}$                          | [174] |
| Perovskite/PCBM                        | —         | $8 \times 10^5$                      | —                                  | [175] |
| Perovskite/P3HT/<br>graphene           | $10^{10}$ | $4.3 \times 10^9$                    | —                                  | [177] |
| Quasi-2D perovskite/<br>InGaZnO        | —         | $>10^5$                              | $5.1 \times 10^{16}$               | [178] |
| Perovskite/AuNPs/<br>FPS-K             | —         | —                                    | $>10^{10}$                         | [179] |
| Perovskite QDs/<br>AuNPs/graphene      | —         | $>10^5$                              | $>10^{13}$                         | [180] |

#### 5.4. Perovskite/other material heterostructures

Apart from the aforementioned materials, many others can also form heterostructures with perovskites to achieve photodetectors with excellent response characteristics. Wei et al. combined flexible quasi-2D perovskites with the metal oxide InGaZnO to fabricate a phototransistor (Fig. 12a) [178]. The capture of photogenerated holes at the heterojunction interface modulated the conductivity of the InGaZnO channel, resulting in a photogating effect. This device achieved an ultrahigh responsivity of  $>10^5 A \cdot W^{-1}$  at 457 nm and a  $D^*$  of  $5.1 \times 10^{16}$  Jones. Additionally, it exhibited a broadband spectral response from 457 nm to 1064 nm. Beyond simple layered structures, researchers have also leveraged the unique properties of nanoparticles to construct heterostructures with perovskites for photodetector fabrication. Kim et al. developed a hybrid structure of Au nanoparticles (AuNPs) and poly (9, 9-bis (4'-sulfonatobutyl)fluorene-alt-1,4-phenylene) potassium (FPS-K), which was then integrated with perovskites to form a heterojunction photodetector (Fig. 12b) [179]. After coupling with AuNPs/FPS-K, the photoluminescence intensity of the perovskite significantly increased due to the additional photoexcited charges generated by the coupling effect between AuNPs and FPS-K. Combined with the trap states at the heterojunction interface that induced a photogating effect, the photocurrent of the perovskite photodetector increased by a factor of  $1.1 \times 10^4$ , achieving an exceptionally high  $D^*$  (Fig. 12c). Similarly, Yadav et al. introduced rhombic dodecahedron Au nanocrystals (RD-AuNCs) into the perovskite QDs/graphene heterostructure to construct a photodetector (Fig. 12d) [180]. Compared to perovskite QDs/graphene heterojunction photodetectors, this structure exhibited significantly enhanced responsivity and  $D^*$  (Fig. 12e).

In conclusion, perovskites, with their unique advantages, have become an important research focus in the field of photodetectors.

Perovskites exhibit a high absorption coefficient, broad spectral response range, and high carrier mobility, making them highly promising for use in photodetectors. By constructing heterostructures with materials such as graphene, TMDs, and organic materials, researchers have successfully introduced the photogating effect, significantly enhancing the gain, responsivity, and  $D^*$  of photodetectors (Table 3). The high light absorption ability of perovskite, combined with the excellent electrical properties of other materials, has enabled perovskite-based photodetectors to demonstrate exceptional performance in various fields. Nevertheless, optimizing material selection and structural design to improve the response speed and sensitivity of the detectors remains a key focus for future research. With the continuous advancement of technology, perovskite-based photodetectors have broad application prospects in fields such as high sensitivity, wide spectral response, and high-speed imaging.

However, there are still some challenges in the application of perovskite materials. Perovskites used in photodetectors exhibit excellent light absorption performance and tunable bandgap characteristics, but their long-term stability remains a critical bottleneck restricting practical device applications [181]. Perovskites are highly sensitive to environmental factors, especially prone to degradation under conditions of humidity, thermal stress, and the presence of oxygen, which leads to rapid deterioration of device performance [182]. Moisture can react with the organic components in perovskites to form hydrate phases, subsequently decomposing into by-products such as PbI<sub>2</sub>; thermally induced ion migration can cause charge accumulation or migration pathways at interfaces or within the bulk, resulting in response drift and even electrical memory effects; oxygen under illumination may also trigger peroxide reactions, accelerating the destruction of the perovskite structure [183,184].

To improve the environmental stability of perovskite photodetectors, commonly used strategies include surface passivation, interface engineering, incorporation of stabilizers, and the adoption of multilayer inorganic or organic-inorganic composite encapsulation structures [185]. Moreover, device design must also address ion migration control, for example, by employing ion-blocking layers or bipolar transport balance structures to suppress the formation of ion-driven electric fields. Therefore, future research should continue focusing on enhancing the structural and performance stability of perovskites under multi-field coupled environments (light, heat, electric field, moisture, oxygen) to facilitate their transition from laboratory prototypes to practical applications [186,187].

## 6. Conclusion and outlook

This paper provides a comprehensive review of the application of the photogating effect in photodetectors and its impact on performance. The photogating effect significantly enhances key performance parameters of photodetectors, such as gain, responsivity, and specific detectivity, by modulating carrier transport characteristics. Studies show that the generation of the photogating effect is primarily related to trap states in the material, which can capture photogenerated carriers and thus influence carrier transport in the conductive channel. By constructing heterojunctions or introducing specific material structures, the photogating effect can be effectively controlled to further optimize photodetector performance. In low-dimensional materials, the modulation of the photogating effect is mainly achieved through surface states and defects in the material. For example, quantum dots, nanowires, and two-dimensional materials exhibit excellent photogating effects due to their limited size and low screening effect. Through the rational design of heterostructures, such as combining quantum dots with metal oxides or two-dimensional materials with graphene, the performance of photodetectors can be significantly improved. Moreover, perovskite materials, due to their high light absorption coefficient and unique optoelectronic properties, have also become an important focus in the study of the photogating effect. By constructing heterojunctions with

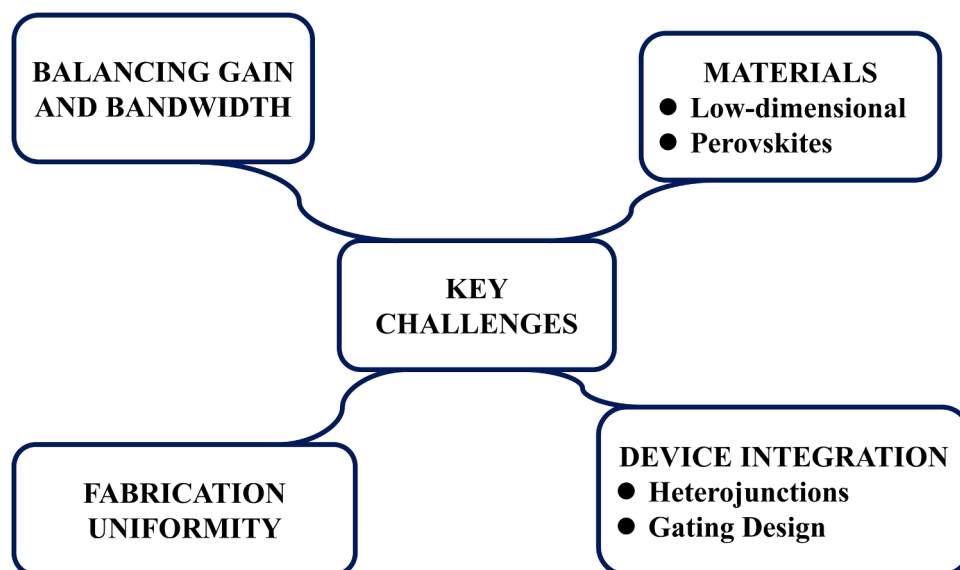


Fig. 13. The key challenges faced in the application of the photogating effect.

materials such as graphene and transition metal dichalcogenides, perovskite-based photodetectors show significant advantages in responsivity, gain, and specific detectivity. This paper also explores application examples of the photogating effect in various photodetectors, including quantum dots, nanowires, two-dimensional materials, and perovskite materials. These research results suggest that the photogating effect not only significantly improves the performance of photodetectors but also provides important theoretical and experimental foundations for designing new high-performance photodetectors. By rationally controlling the photogating effect, the performance of photodetectors can be optimized to meet the needs of various application scenarios.

Although significant progress has been made in applying the photogating effect in photodetectors, there are still challenges and future directions for development. First, the control of the photogating effect needs further optimization to achieve higher gain and faster response speed. This requires in-depth research into the trap state characteristics of materials and how to control these trap states through material design and structural optimization. For example, precisely controlling the defect and impurity distribution in materials can enable fine-tuning of the trap states, thereby optimizing the photogating effect. Although the photogating effect has been widely observed in various low-dimensional materials and perovskite photodetectors and is closely related to defect states, current studies on defect states remain primarily qualitative. Most works infer the existence and types of trap states indirectly based on macroscopic electrical behaviors of devices, such as transfer characteristics and photoresponse delays. However, key parameters of defect states—such as energy level positions, density distributions, and capture cross sections still lack systematic quantitative characterization and statistical data. This limitation hinders the in-depth understanding and controllable design of the photogating effect. Therefore, future research urgently needs to incorporate advanced quantitative characterization techniques, such as deep-level transient spectroscopy (DLTS), photo-thermal deflection spectroscopy (PDS), photo-induced current transient spectroscopy (PICTS), and time-resolved photoluminescence (TRPL), to achieve precise identification and modeling of defect states. Second, the application of the photogating effect in new materials still requires further exploration. With the rapid development of nanotechnology and materials science, new low-dimensional and composite materials are emerging. These materials possess unique optoelectronic properties that may offer new ideas and methods for the study of the photogating effect. For instance, the interlayer coupling of two-dimensional materials and

defect engineering in perovskite materials could provide new pathways for controlling the photogating effect. In addition, the application of the photogating effect within single perovskite materials requires further investigation. Regulating defect states in pure perovskite devices to modulate the photogating effect remains a major challenge for the application of the photogating effect in such systems. The presence of the photogating effect may promote the utilization of perovskite materials in various fields, such as solar cells. Third, the stability of the photogating effect in practical applications needs to be further improved. In real-world use, photodetectors need to maintain stable performance under various environmental conditions. Therefore, research into improving the stability and reliability of the photogating effect is one of the key future directions. This may involve encapsulation techniques for materials, interface engineering, and long-term stability testing of devices. Finally, the photogating effect has vast potential in the design of new photodetectors. As the demand for high-performance photodetectors increases, the photogating effect is expected to be applied in more fields. For example, in biomedical imaging, environmental monitoring, and communication technologies, the photogating effect can significantly improve the performance of photodetectors to meet the requirements of different application scenarios. Future research can further explore the specific applications of the photogating effect in these fields, developing photodetectors with higher performance and broader applicability. As of now, the key challenges faced in the application of the photogating effect are shown in Fig. 13.

In conclusion, significant progress has been made in applying the photogating effect in photodetectors, but many challenges remain to be addressed. By deeply studying material properties, optimizing structural designs, and improving stability, the photogating effect is expected to play an even greater role in the future of photodetector technology, driving the development of related fields.

#### CRediT authorship contribution statement

**Hao Liu:** Writing – original draft. **Zihan Yi:** Writing – review & editing. **Cong Chen:** Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



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## Data availability

No data was used for the research described in the article.

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